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ORIGINAL PAPER

Robust design of a sustainable and resilient bioethanol supply chain under operational and disruption risks

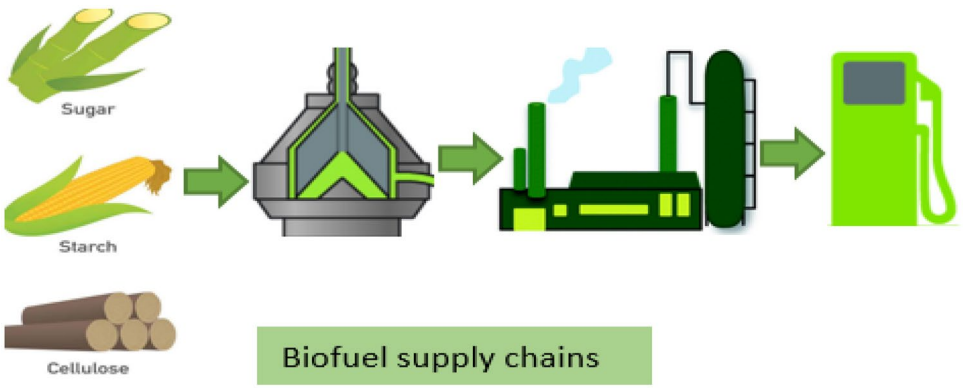
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Abstract

In this work, we propose a mixed-integer linear programming model to address the design and planning of a multi-feedstock lignocellulosic bioethanol supply chain network. In order to provide resiliency against existing epistemic uncertainties and disruption risks in the supply chain, a hybrid robust stochastic-possibilistic programming approach is employed. The proposed model minimizes total expected cost of the supply chain over non-disruption and disruption scenarios while limiting greenhouse gas emissions for sustainability considerations. The model determines the optimal supply chain strategic and tactical decision variables such as location, capacity and technology of biorefineries, transportation modes, shipments, inventory levels and production and import amounts. The performance of the model is evaluated through a real case study developed in Iran. Comparing the proposed resilient model with its non-resilient counterpart reveals that the proposed model is superior both in cost and GHG emissions reductions.

Graphic abstract



Keywords Bioethanol supply chain · Lignocellulosic biomass · Resiliency · Disruption · Robust possibilistic programming · Stochastic programming

Introduction

Renewable sources of energy, such as solar, wind, biomass and geothermal have attracted increasing attention because of growing energy demand and seeking environmentally sustainable sources of energy (Bairamzadeh et al. 2015). Among the renewable sources of energy, biomass is of particular importance because of its widespread availability and its potential applicability as a sustainable source

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of energy and material (Pérez et al. 2017). The European Union defines biomass as the biodegradable elements of the products, urban and industrial waste, and agricultural and forestry residues which are used as the feedstock in order to produce biofuels and generate heat and power in the bioenergy production systems (Ghaderi et al. 2016). Various kinds of biofuels are produced from biomass feedstocks such as bioethanol, biodiesel, biooil and biogas (Sharma et al. 2013). Biofuels can be substituted or blended with fossil fuels in several ways. Therefore, many countries have set biofuel targets in their future energy plans. Iran has set a target of 5% blending of bioethanol and biodiesel with gasoline and diesel by the year 2025 ("National Document for Knowledge-based Development of Renewable energies" 2015).

The biomass feedstocks, biofuels and the related production processes are categorized into four generations. The feedstock of the first-generation biofuels is mainly obtained from some edible resources such as starch, sugars, animal fat and vegetable oils. Although the production processes are well developed and commercialized, the first-generation biofuels conflict with food supply chain. The second-generation biofuels are produced from non-edible feedstock including lignocellulosic biomass, agricultural and forestry residues, and municipal and industrial waste. The third-generation biofuel, recently developed, is obtained from algae. The disadvantage of the second- and third-generation biofuels is their difficult and complex production processes (Ghaderi et al. 2016). But they are preferred since they use readily available and cheap raw materials with no competition with food and reduce greenhouse gas (GHG) emissions (Awudu and Zhang 2012). The fourth-generation biofuel, still not matured, is tried to be produced through genetically modifying or synthetizing microorganisms (Zandi Atashbar et al. 2017).

Along biofuel production, biomass feedstock flows from the supply sites to the demand zones, passing through some facilities and undergoing different processes called the biomass supply chain (Ghaderi et al. 2016). The operations of biofuel supply chains can be separated into three segments. The upstream segment covers the activities from biomass production to biomass delivery to biorefineries including biomass production, harvest, collection, pre-treatment and storage. The midstream segment only encompasses biomass conversion to biofuel at biorefineries. The downstream segment involves biofuel storage and distribution to customers (De Meyer et al. 2014). Harvesting and collection are usually done in a limited time at biomass production site (Ba et al. 2016). There are different biomass collection activities such as loafing, baling, wet chopping, dry chopping, single-pass harvesting, multi-pass harvesting and whole-crop harvesting (Zandi Atashbar et al. 2017). Storage is required to lessen dry matter loss and protect biomass (Gold and Seuring 2011). It can be open air or roof covered and air fan. It can be done on the farms, in centralized storage sites or before the processes in conversion facilities site (Ba et al. 2016). Preprocessing is necessary to reduce degradations, improve handling, prepare biomass for the conversion processes and decrease transportation costs through densifying biomass (Zandi Atashbar et al. 2017). There are some common types of preprocessing activities such as ensiling, drying, pelletization and torrefaction (Gold and Seuring 2011). Broadly speaking, biomass can be converted into biofuel at biomass conversion facilities by two major categories of technologies, namely biochemical and thermochemical (Yue et al. 2014). Biochemical technology employs hydrolysis and fermentation processes, and thermochemical technology employs gasification and pyrolysis processes in biofuel production (Parajuli et al. 2015). Biomass and biofuel can be transported via several transportation modes such as roadway, railway and waterway. Depending on the type of biomass, shape, distance, accessibility and the demand of customers, one transportation mode or a combination of them can be selected (Axelsson et al. 2012). Special characteristics of biomass such as wide collection territory, degradation (Ba et al. 2016), seasonality and weather-related variability, high moisture content, low energy content and low bulk density distinguish biomass supply chain from other supply chains (Iakovou et al. 2010).

Bioethanol, a type of liquid biofuel, can be blended into gasoline up to 10% without the need for a major change in the engine of vehicles (Dufey 2006). It is derived from a number of biomass feedstock types mainly categorized as edible and non-edible (or lignocellulosic) ones (Muktham et al. 2016). Main edible biomass feedstock types include sugarcane, sugar beet, corn, wheat and cassava. And common non-edible feedstock types are grasses, straw, stovers, wood and agricultural residues (Ho et al. 2014). The production of biofuels from edible biomass feedstock endangers food supply. However, the production of bioethanol from lignocellulosic biomass has challenges such as low energy and bulky volume of biomass, wide geographically distributed supply network of biomass and seasonality of biomass supply (Maheshwari et al. 2017). These challenges highlight the importance of designing an economic lignocellulosic biofuel supply chain (LBSC). On the other hand, since environmental concerns are one of the main reasons for

moving toward biofuels, it seems essential to include environmental sustainability in design of biofuel supply chain networks (BSCNs). Environmental sustainability is mainly considered as minimizing GHG emissions in design of most BSCN models (Ghaderi et al. 2016). Besides, Biofuel supply chains (BSCs) are coupled with uncertainty due to their complex and dynamic nature. The most common sources of uncertainty include biomass supply, biofuel demand, different prices and costs, and technology. Uncertainties have greater impacts on the efficiency of supply chains in strategic level designs much more than tactical-/operational-level designs (Balaman and Selim 2015). Uncertainties could be classified into two major categories: (1) randomness uncertainty because of the random nature of events and (2) epistemic uncertainty caused by lack of knowledge about events (Mula et al. 2007).This form of uncertainty could be due to limited and inaccurate data, measurement error, incomplete knowledge, limited understanding, imperfect models, subjective judgment, ambiguities, etc. (Walker et al. 2003). Due to insufficient knowledge, probability distribution of parameters with epistemic uncertainty cannot be extracted (Bairamzadeh et al. 2018), and these parameters are modeled by fuzzy intervals in the setting of possibility theory (Mula et al. 2007). Stochastic and possibilistic programming approaches are broadly applied to handle these two types of uncertainties, respectively (Pishvaee and Torabi 2010). These approaches do not control robustness of the optimal solution (Ghaderi et al. 2018). Therefore, robust scenario-based stochastic programming and robust possibilistic programming (RPP) approaches have been presented to control feasibility and/or optimality robustness of the optimal solution (Bairamzadeh et al. 2018).

On the other hand, the increasing occurrence of natural disasters, wars, financial crises, etc., indicates the necessity of designing supply chains resilient to disruptions (disruption risks) in addition to uncertainties (operational risks). Disruption risks refer to uncertainties arisen from disruptions caused by natural disasters such as flood, drought and hurricane, as well as man-made disasters such as wars, terrorist attacks, financial crisis and strikes. Disruption risks although happen with a less frequency compared with operational risks, have greater impacts (Tang 2006). Disruption risks could also be categorized into site-dependent and site-independent ones. Natural disasters are mostly site-dependent, and man-made disasters are mainly site-independent disasters. Like other supply chains, disruption risks can adversely impact the performance of BSCNs. They can have impacts on biomass supply, biofuel production and demand, material flow, etc. Therefore, it seems important to design a biofuel supply chain network resilient to both operational and disruption risks.

In the literature, several definitions have been given for supply chain resilience (SCR). Christopher and Peck (2004) were the first ones who defined SCR based on ecosystem definition of resilience, as the capability of a supply chain to come back to its initial state or to go to a new, more desirable state after experiencing a disruption. Ponomarov and Holcomb (2009), considering multidisciplinary perspectives, developed definition of SCR as the compatibility of a supply chain to prepare for sudden disruptions, react to disturbances and recover from them through continuing the operations at the acceptable level of connection and keeping control over structure and performance. Kamalahmadi and Parast (2016) defined supply chain resilience as the compatibility of a supply chain to reduce the probability of confronting unpredicted events, prevent the deployment of disruptions through keeping control over structure and performance, and react to and recover from disturbances by prompt, effective and predetermined plans to overcome the disruption and return the supply chain to a robust state.

There are several other definitions for SCR, indicating lack of consensus (Tukamuhabwa et al. 2015). According to the different definitions of SCR and by reviewing the related literature, the studies mathematically accounting for SCR, especially in strategic level design, can be categorized into two main groups. The first group of studies puts its focus on designing supply chain networks protected against disruptions. Expected value approach, value-at-risk (VaR) and conditional value-at-risk (CVaR) measures have been broadly utilized in these studies to optimize design of resilient supply chains. Worst-case approaches and robustness measures have also been applied in these papers. The second group of studies tries to design supply chain networks with the ability of recovering after they have experienced disruptions. This group applies measures such as restoration cost, recovery time and actual demand met for optimizing design of supply chains (Fahimnia and Jabbarzadeh 2016).

The purpose of this study is to present a relatively comprehensive model that shows decision makers (DMs) a way to move toward cleaner production with less greenhouse gas emissions. To do so, a mixed-integer linear programming (MILP) model is designed to optimize a sustainable multi-period multi-feedstock bioethanol supply chain network. A hybrid robust stochastic-possibilistic programming (RSPP)

approach is used to handle biomass supply disruption in addition to the epistemic uncertainty of different parameters of the model. A real-world case study in Iran is implemented to evaluate the validity and performance of the proposed model. Besides, the presented robust and resilient model is compared with its non-resilient counterpart and some valuable managerial implications are achieved.

The following particular features distinguish this study from other studies in the literature:

- The proposed model accounts for disruption risks in addition to epistemic operational risks while considering environmental sustainability.
- The proposed robust stochastic-possibilistic programming (RSPP) approach provides resiliency and robustness against operational and disruption risks. It also provides more flexibility for DMs by taking the advantage of fuzzy *Me* measure. To the best of our knowledge, this is the first study in the literature which uses RSPP approach for the design of a biofuel supply chain network (BSCN).
- A real case study is conducted to assess the reliability and efficiency of the proposed model.

The remainder of this paper is structured as follows. Section 2 reviews the related literature. Section 3 provides the problem definition and presents the mathematical model. Section 4 overviews RPP approach and introduces the proposed robust stochastic-possibilistic programming (RSPP) model. Section 5 provides an execution of the model using a real case study. Section 6 concludes the paper and discusses possible future research.

Literature review

In this section, we review the studies on BSCN design which have considered operational and/or disruption risks. First, we go over the studies in which the operational risks (uncertainties) have been included. Since there are many papers on this subject, we review some recent papers. Then we investigate the studies on BSCN design in which disruption risks have been taken into account. These studies are very limited in the literature.

Uncertainties have been frequently considered in the design of BSCNs to make the models more realistic and reliable. Authors have taken into consideration uncertainty of parameters such as biomass supply, biomass quality, biofuel demand, different prices and costs, and environmental impacts. Osmani and Zhang (2017) presented a two-stage stochastic MILP model for design of a multi-feedstock bioethanol supply chain network. Their multi-objective sustainable model accounted for the uncertainty of bioethanol price and demand. d'Amore and Bezzo (2017) developed an MILP model to optimize the profitability of a biomass-based ethanol and electricity production. They applied regret theory to handle the uncertainty of technology conversion efficiency. Xie and Huang (2018) presented a multistage stochastic MILP model to formulate the evolving uncertainty of biofuel demand in a biofuel supply chain. In order to solve the model, they transformed it to a two-stage model and then applied an enhanced nested decomposition method. Ghelichi et al. (2018) designed a Jatropha-to-biodiesel supply chain network and applied an extended two-stage stochastic programming model to capture uncertainties of biodiesel demand and biomass supply. They introduced min–max relative regret approach in a soft-worst-case framework to solve their model. Asadi et al. (2018) designed a bi-objective sustainable algae BSCN model and considered uncertainty of biofuel demand and delivery times. For solving the formulated NP-hard model, multi-objective particle swarm optimization (MOPSO) and non-dominated sorting genetic algorithm II (NSGA-II) were applied. Quddus et al. (2018) introduced a two-stage chance-constrained stochastic programming model to cope with the uncertainties due to feedstock seasonality in a BSCN. They applied a combined sample average approximation algorithm to solve their model. Ye et al. (2018) designed a risk-sharing model to coordinate the decision-making behavior and to help the decision makers in a cassava-based bioethanol supply chain to understand how to increase the supply chain resiliency under yield and demand uncertainty risk.

Recently, robust programming has been used to provide robustness in designing BSCNs. Authors use robust convex programming, robust possibilistic programming (RPP) and robust scenario-based stochastic programming for deep, epistemic and randomness uncertainties, respectively. For instance, Bairamzadeh et al. (2015) presented an RPP approach to optimize the design of a multi-objective LBSC network under the uncertainty of bioethanol demand, selling prices and environmental impacts. Yue and You (2016) developed a nested stochastic robust optimization model to handle both strategic and operational supply chain uncertainties. The resulting multi-level MILP model was solved by a decomposition-based column-and-constraint generation

algorithm. Employing possibilistic mean value and absolute deviation of fuzzy numbers, Babazadeh et al. (2017) introduced a new possibilistic programming approach and RPP approach to design a sustainable biodiesel supply chain. Zhang and Jiang (2017) proposed a robust MILP model to design a waste cooking oil-to-biodiesel supply chain network under biodiesel price uncertainty. Bairamzadeh et al. (2018) developed a hybrid robust optimization model to capture different kinds of uncertainty in a multi-feedstock LBSC. Their hybrid robust optimization model included robust convex programming, robust possibilistic programming and robust scenario-based stochastic programming to handle deep, epistemic and randomness uncertainties of parameters, respectively. Ghaderi et al. (2018) proposed an RPP model to optimize the design of a multi-objective sustainable bioethanol supply chain network. They considered several uncertain parameters in their model constraints and economic, environmental and social objectives. Babazadeh (2018) designed an MILP model for a biodiesel supply chain network. He applied set-induced robust optimization method to handle the deep uncertainty of the supply chain parameters including feedstock supply, biodiesel demand, and different costs and incomes. Mousavi et al. (2018) presented a multi-objective hybrid RPP model for the design and planning of a multi-period multi-feedstock LBSC network considering uncertainty of various parameters in all three economic, environmental, and social objective functions and constraints.

Although operational risks have been frequently considered in the design of BSCNs, disruption risks have not gained enough attention and there are really few studies that have accounted for disruption risks. For example, Huang and Pang (2013) incorporated infrastructure resilience assessments into a bioethanol design model by developing a bi-objective stochastic MILP model. The objective functions were minimization of total expected cost of the BSC and minimization of expected cost of resilience against potential seismic hazards across all scenarios. Marufuzzaman et al. (2014) proposed a model for the design of a reliable multimodal logistics network for a BSC and investigated the impact of intermodal hub-related risk on this network. To solve the proposed NP-hard model, they developed a customized solution approach based on the traditional Benders decomposition with stronger cuts. Azadeh et al. (2014) designed a scenario-based stochastic model for optimization of a multi-period, multi-feedstock and multi-commodity BSC. They also proposed a robust model which only provided optimality robustness (and not feasibility robustness) against uncertainties of biofuels prices and demands. They also included possible disruptions of supply resources and the connecting links of the network in their model. Bai et al. (2015) analyzed the impact of both site-independent and site-dependent disruptions at biorefinery locations of a bioethanol supply chain. They applied discrete and continuous reliable facility location models for the design of a reliable bioethanol supply chain. They solved the discrete model by a customized Lagrangian relaxation algorithm and the continuous model by a continuous approximation method. Azadeh and Arani (2016) applied a hybrid system dynamics-mathematical programming approach in designing and planning a biodiesel supply chain. In the first stage, some parameters of the BSCN model were simulated in the system dynamics model, considering water resource and land limitations and technological issues. Then, the estimated parameters were used as inputs to a stochastic MILP model which also considered disruption in links among biomass fields, biorefineries and biofuel consumption markets. Finally, they proposed a robust counterpart of their model which only accounted for optimality robustness. Poudel et al. (2016) determined the optimal locations for multimodal facilities and biorefineries, production planning and material flow by developing a mixed-integer nonlinear programming (MINLP) model. They also offered suggestions on determining which links' reliability can be improved under specific budget limitations to prevent possible losses due to disruption of transportation links. Marufuzzaman and Ekşioğlu (2016) designed a cost-efficient and reliable BSCN. They considered a multimodal transportation network to hedge against biomass supply fluctuations and natural disasters. They developed an accelerated Benders decomposition algorithm and a hybrid rolling horizon algorithm to solve their MINLP model. Maheshwari et al. (2017) proposed an MILP model for the design and planning of a BSCN. The objective function was the sum of the total cost during the non-disruption and disruption scenarios weighted by their relevant probability of occurrence which also measured the expected disruption cost (EDC) of the BSC. Their model determined the location and capacity of regional biomass preprocessing depots (RBPDs) and biorefineries. Liu et al. (2017) designed a reliable supply chain network for biomass collection against feedstock seasonality and time-varying disruption. They captured both seasonality and disruptions risks by a set of chance constraints. Poudel et al. (2018) presented an MINLP model to design a BSCN considering site-dependent facility failure and congestion. Their model determined the optimal

Table 1 Recent BSCND papers that consider disruption risks

Reference	Source of disruption risk	Objective function			Location/ allocation	Capacity of facilities	Type of technology	Flow of material	Inventory planning	Transportation mode	Import/ export of biomass/ biofuel	Modeling approach	Sustainability issues	Source of operational risk					Disruption modeling approach
		Min total expected cost	Min total expected resilience cost	Max expected profit										Biomass supply	Biofuel demand	Biomass/ biofuel price	Costs	Multi-period	
Huang and Pang (2013)	Biorefineries	*	*		*	*		*				MILP							Two-stage stochastic programming
Marufuzzaman et al. (2014)	Intermodal hubs	*			*	*		*		*		MILP							Stochastic programming
Azadeh et al. (2014)	Links, biomass fields			*	*	*	*	*	*			MILP			*	*		*	Two-stage stochastic programming, Scenario-based robust programming
Bai et al. (2015)	Biorefineries	*			*							MINLP							Stochastic programming
Azadeh and Arani (2016)	Links between biomass fields, biorefineries and markets			*	*	*	*	*	*			MILP, System dynamics	*	*	*	*		*	Scenario-based stochastic programming, scenario-based robust programming

Table 1 (continued)

Reference	Source of disruption risk	Objective function			Location/ allocation	Capacity of facilities	Type of technology	Flow of material	Inventory planning	Transportation mode	Import/ export of biomass/ biofuel	Modeling approach	Sustainability issues	Source of operational risk					Disruption modeling approach
		Min total expected cost	Min total expected resilience cost	Max expected profit										Biomass supply	Biofuel demand	Biomass/ biofuel price	Costs	Multi-period	
Poudel et al. (2016)	Links between hubs and biorefineries	*			*	*		*				MINLP							Stochastic programming
Marufuzzaman and Ekşioğlu (2016)	Intermodal hubs	*			*	*		*	*	*		MINLP						*	Stochastic programming
Maheshwari et al. (2017)	Biomass supply locations	*			*	*		*	*			MILP						*	Scenario-based stochastic programming
Liu et al. (2017)	Collection facilities	*			*			*	*			MINLP						*	Possibilistic programming
Poudel et al. (2018)	Multimodal facilities	*			*	*		*	*			MINLP						*	Stochastic programming
Fattahi and Govindan (2018)	Storage facilities	*			*	*		*	*	*		MINLP	*	*				*	Multistage stochastic programming
This study	Biomass supply zones	*			*	*	*	*	*	*	*	MILP	*	*	*	*	*	*	Robust stochastic-possibilistic programming

location of biorefineries and multimodal facilities, production, storage, and routing plans. They solved their NP-hard model using a hybrid constraint generation-based rolling horizon algorithm. Fattahi and Govindan (2018) proposed a multistage stochastic programming model for the design and planning of a BSCN considering disruption risk, feedstock seasonality and uncertainty, and sustainability. They included environmental and social aspects of sustainability in constraints of their model. They generated scenarios using Monte Carlo simulation and utilized scenario tree construction and scenario reduction techniques. Benders decomposition algorithm and rolling horizon algorithm were applied to solve the proposed MINLP model.

Table 1 reports the features of the papers that have considered disruption in the design of BSCNs. It also shows the characteristics of this study in comparison with other relevant studies. Observing the studies in Table 1 indicates that there are few studies that have considered both operational and disruption risks as well as sustainability considerations in design of BSCs. To the best of our knowledge, this is the first study in the literature of BSCN design which accounts for disruptions in addition to epistemic uncertainties. Besides, this is the first study which puts its effort to design a robust and resilient BSC.

Problem description and formulation

Problem statement

In this study, a multi-period multi-feedstock LBSC network is designed under operational and disruption risks. Figure 1 illustrates a general structure of the LBSC network under study. Biomass feedstock is harvested in supply zones, and then, the harvested biomass is shipped to biorefineries. Biochemical or thermochemical technologies applied in biorefineries convert biomass into bioethanol which produce by-products as well. By-products of these two technologies are bioelectricity and mixed alcohols, respectively. Bioethanol is then shipped to demand zones, oil refineries in this study, and there it is blended into gasoline. Both biomass and bioethanol could be shipped via road and rail transportation modes across the network. Import of limited quantities of bioethanol to demand zones is allowed if it is needed to meet the desired demand or even if economically justified.

The proposed model minimizes total cost of the BSC under study as economic objective function. Both strategic and tactical decision variables will be optimized through optimizing the economic performance of the model. Strategic decision variables include location, production technology and processing capacity of biorefineries. Besides, the model determines the following tactical decisions in each planning period and under each scenario:

- Quantity of harvested biomass of each supply zone, sent to each biorefinery;
- Quantity of each type of biomass, processed by each biorefinery;
- Quantity of produced bioethanol, sent to each oil refinery;
- Quantity of biomass inventory level at each biorefinery;
- Quantity of bioethanol inventory level at each biorefinery; and
- Quantity of bioethanol imported to each oil refinery.

The following parameters are epistemic sources of uncertainty because of limited available data and lack of knowledge:

- Biomass feedstock yield because of variations in rainfall and other unpredictable weather conditions;
- Bioethanol demand of oil refineries due to variations in their gasoline production; and
- Several prices and costs caused by market fluctuations.

This study also takes into account the disruption in biomass feedstock yield due to possible drought scenarios.

Optimization model

The mentioned problem is formulated as an MILP model. In this section, the model nomenclature, objective function and constraints are described in detail.

Nomenclature

The indices, parameters and decision variables of the model are introduced in this section. The uncertain parameters are denoted by symbols with a tilde (~).

Indices	
b	Biomass feedstock types, $b=1, \ldots, B$
c	By-products, $c=1, \ldots, C$
i	Biomass supply zones, $i=1, \ldots, I$
j	Candidate biorefinery locations, $j=1, \ldots, J$
k	Bioethanol demand zones, $k=1, \ldots, K$
m	Transportation modes, $m=1, \ldots, M$

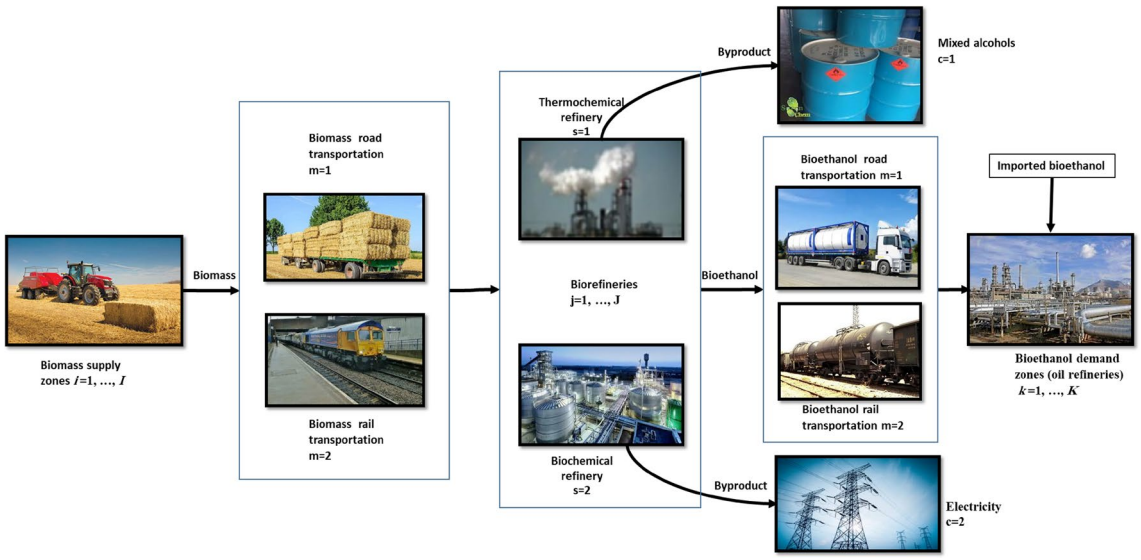


Fig. 1 General structure of the studied BSCN

n	Scenarios including non-disruption scenario, n = 1, …, N
s	Conversion technologies, $s = 1, \ldots, S$
t	Planning periods, $t = 1, \ldots, T$
Parameters	
bf_t	Blending fraction of bioethanol into gasoline in planning period t
$\widetilde{\mathrm{bp}}_{c,t}$	Sale price of by-product c in planning period t, \$/gal or \$/MWh
chm_b	Carbon emissions of harvesting biomass type b, ton CO_2-equiv/ton
$\mathrm{cpm}_{b,s}$	Carbon emissions of processing biomass type b with technology s, ton CO_2-equiv/ton
cte_m	Carbon emissions of transporting bioethanol by transportation mode m, ton CO_2-equiv/gal/km
$\mathrm{ctm}_{b,m}$	Carbon emissions of transporting biomass type b by transportation mode m, ton CO_2-equiv/ton/km
$\mathrm{disrd}_{j,k,m}$	Distance between biorefinery j and demand zone k by transportation mode m, km
$\mathrm{dissr}_{i,j,m}$	Distance between supply zone i and biorefinery j by transportation mode m, km
$\widetilde{\mathrm{fcr}}_{j,s}$	Seasonalized fixed setup cost of biorefinery j with technology s, \$
$\widetilde{\mathrm{hcm}}_{b,i}$	Harvesting cost of biomass type b in supply zone i, \$/ton
$\widetilde{\mathrm{ice}}_{j,t}$	Inventory holding cost of bioethanol in biorefinery j at time period t, \$/gal/season
$\widetilde{\mathrm{icm}}_{b,j,t}$	Inventory holding cost of biomass feedstock type b in biorefinery j at time period t, \$/ton/season
$\widetilde{\mathrm{imp}}_t$	Import cost of bioethanol in planning period t, \$/gal
$l_{i,b}$	Land area of supply zone i allocated for cultivating biomass type b, ha
$\widetilde{\mathrm{lce}}_{j,k,m}$	Truck loading and unloading cost of bioethanol by transportation Mode m from biorefinery j to demand zone k, \$
$\widetilde{\mathrm{lcm}}_{b,i,j,m}$	Truck loading and unloading cost of biomass type b by transportation mode m from supply zone i to biorefinery j, \$
lr_s	Minimum amount of biomass that can be processed by refineries with technology s, ton/season
mc_b	Moisture content of biomass type b
p_n	Probability of occurrence of each scenario n
$\widetilde{\mathrm{pce}}_{j,s}$	Production cost of bioethanol in biorefinery j by technology s, \$/gal
$\widetilde{\mathrm{pg}}_{k,t}$	Amount of gasoline produced in each bioethanol demand zone (oil biorefinery) k in planning period t, ton/season
tcd_m	Capacity of transportation mode m, dry ton

$\widetilde{\text{tce}}_{j,k,m}$	Transportation cost of bioethanol by transportation mode m per unit of distance between biorefinery j and demand zone k, \$/km
tcl_m	Liquid capacity of transportation mode m, gal
$\widetilde{\text{tcm}}_{b,i,j,m}$	Transportation cost of biomass type b by transportation mode m per unit of distance between supply zone i and biorefinery j, \$/km
uc	Maximum allowed amount of carbon to be emitted during the planning horizon, ton CO_2-equiv
uim_t	Maximum allowed amount of bioethanol to be imported during planning period t, gal/season
ur_s	Maximum amount of biomass that can be processed by refineries with technology s, ton/season
$\widetilde{\text{vcr}}_{j,s}$	Variable setup cost of biorefinery j with technology s, \$/ton
$\widetilde{\text{ym}}_{b,i,t,n}$	Yield of biomass type b of supply zone i in planning period t under scenario n, ton/ha/season
α_n	Minimum confidence level of supply chance constraint under scenario n
β_n	Minimum confidence level of demand chance constraint under scenario n
γ	Optimistic–pessimistic parameter
δ_1	Unit penalty cost of violation from supply chance constraint
δ_2	Unit penalty cost of violation from demand chance constraint
$\theta_{b,t}$	Deterioration rate of biomass type b during storage in time period t
$\lambda_{b,s,c}$	Conversion factor of biomass type b through technology s to by-product c, gal/dry ton or MWh/dry ton
σ	Importance weight of optimality robustness under randomness of parameters
ν	Maximum ratio of crop residues that is allowed to be removed from supply zones
$\varphi_{b,s}$	Conversion factor of biomass type b through technology s to bioethanol gal/dry ton
ϕ	Importance weight of optimality robustness under fuzziness of parameters
ψ	Minimum efficiency percentage that is acceptable for biorefinery capacity
$\omega_{b,i,t}$	Harvesting loss fraction of biomass type b in supply zone i in planning period t
Binary variables	
$Y_{j,s,t}$	1 if a biorefinery with technology s is setup at location j in planning period t and 0 otherwise
Continuous variables	
$\text{CM}_{j,s,t,n}$	Amount of biomass type b processed in biorefinery j through technology s in planning period t under scenario n, ton
$\text{CR}_{j,s,t}$	Biomass processing capacity of biorefinery j with technology s in planning period t, ton
$\text{HM}_{b,i,t,n}$	Amount of biomass type b harvested from supply zone i in planning period t under scenario n, ton
$\text{IE}_{j,s,t,n}$	Inventory level of bioethanol in biorefinery j with technology s during time period t under scenario n, gal
$\text{IM}_{b,j,t,n}$	Inventory level of biomass type b in biorefinery j during time period t under scenario n, ton
$\text{IME}_{k,t,n}$	Amount of bioethanol imported to demand zone k in planning period t under scenario n, gal
$\text{PB}_{c,j,s,t,n}$	Amount of by-product c produced by biorefinery j with technology s in planning period t under scenario n, gal or MWh
$\text{PE}_{j,s,t,n}$	Amount of bioethanol produced by biorefinery j with technology s in planning period t under scenario n, gal
$\text{TE}_{j,s,k,m,t,n}$	Amount of bioethanol transported from biorefinery j with technology s to demand zone k by transportation mode m in planning period t under scenario n, gal
$\text{TM}_{b,i,j,m,t,n}$	Amount of biomass type b transported from supply zone i to biorefinery j by transportation mode m in planning period t under scenario n, ton
ζ_n	Variable used for linearization absolute function

Objective function

The objective of the optimization model is to determine the optimal supply chain strategic and tactical decision variables. To do so, total cost of the supply chain is minimized over all scenarios, i.e., the scenario without any disruption and disruption scenarios. Total cost (Z) could be considered as the weighted value of all scenarios, where the weights are

the probability of occurrence of scenarios. Therefore, the objective function can be written as Eq. (1):

$$\min Z = \left(1 - \sum p\right) Z_{\mathrm{ND}} + \sum p Z_{\mathrm{D}} \tag{1}$$

where p is the probability of occurrence of each disruption scenario. Thus, $1 - \Sigma p$ becomes the probability of occurrence of the scenario without any disruption. Also, z_{ND} is the total cost of BSC without any disruption and z_{D} is the total cost of BSC with disruption.

$$\min Z = Z_{\mathrm{ND}} + \sum p(Z_{\mathrm{D}} - Z_{\mathrm{ND}}) = Z_{\mathrm{ND}} + \mathrm{EDC} \tag{2}$$

Second term of the objective function in Eq. (2) can be interpreted as the expected increase in total cost due to disruption scenarios. Therefore, EDC is the expected disruption cost (Maheshwari et al. 2017).

The objective function minimizes total cost minus sale revenue from by-products. Thus, z_{ND} could be stated as follows:

Z_{ND} = fixed setup cost of biorefineries + variable setup cost of biorefineries + harvesting cost of biomass + production cost of bioethanol + transportation cost of biomass from supply zones to biorefineries + transportation cost of bioethanol from biorefineries to demand zones + inventory cost of biomass + inventory cost of bioethanol + import cost of bioethanol − sale revenue from by-products. It should be noted that since by-products are obtained at no extra cost during the bioethanol production process and could be sold at biorefineries gate, the revenue from by-products sale is considered as a side income of bioethanol supply chain. Therefore, it could be subtracted form total cost of bioethanol supply chain.

Total cost (Z) can be formulated as Eq. (3). Setup (investment) cost of biorefineries is broken down into fixed setup cost and capacity (variable) setup cost, in order to account for economy of scale in capacity investment (Osmani and Zhang 2014). Setup cost of biorefineries consists of a fixed part, not depending on biorefineries capacity, and a variable part, depending on their capacity. Variable cost could be obtained by multiplying capacity of biorefineries by unit variable setup cost (the second term). Harvesting cost of biomass, production cost of bioethanol, import cost of bioethanol and inventory costs of biomass and bioethanol (the third, fourth, seventh, eighth, and ninth terms) are simply calculated by multiplying amount of harvested biomass, amount of produced bioethanol, amount of imported bioethanol, biomass and bioethanol inventory levels by their corresponding unit costs. Transportation cost of biomass, the fifth term, is calculated by multiplying the amount of transported biomass by unit transportation cost between supply zones and biorefineries. Unit transportation cost is obtained through dividing total transportation cost by the capacity. Total transportation cost itself is the summation of transportation cost and truck loading and unloading cost. Moisture content is included to consider the wetness of the transported biomass. Transportation cost of bioethanol, the sixth term, is calculated in likewise. Finally, the last term, i.e., sale revenue from by-products, is the product of produced amount of by-products and their sale prices. The mentioned terms are weighted over all scenarios, except for fixed and variable setup costs of biorefineries, which remain the same in all non-disruption and disruption scenarios.

$$\begin{aligned}
\min Z = {} & \sum_j \sum_s \sum_t \widetilde{\mathrm{fcr}}_{j,s} Y_{j,s,t} + \sum_j \sum_s \sum_t \widetilde{\mathrm{vcr}}_{j,s} CR_{j,s,t} + \sum_n p_n \Bigg(\sum_i \sum_b \sum_t \widetilde{\mathrm{hcm}}_{b,i} \mathrm{HM}_{b,i,t,n} \\
& + \sum_j \sum_s \sum_t \widetilde{\mathrm{pce}}_{j,s} \mathrm{PE}_{j,s,t,n} + \sum_b \sum_i \sum_j \sum_m \sum_t \left(\frac{\mathrm{dissr}_{i,j,m} \widetilde{\mathrm{tcm}}_{b,i,j,m} + \widetilde{\mathrm{lcm}}_{b,i,j,m}}{\mathrm{tcd}_m} \right) \left(\frac{1}{1 - \mathrm{mc}_b} \right) \mathrm{TM}_{b,i,j,m,t,n} \\
& + \sum_j \sum_s \sum_k \sum_m \sum_t \left(\frac{\mathrm{disrd}_{j,k,m} \widetilde{\mathrm{tce}}_{j,k,m} + \widetilde{\mathrm{lce}}_{j,k,m}}{\mathrm{tcl}_m} \right) \mathrm{TE}_{j,s,k,m,t,n} + \sum_k \sum_t \widetilde{\mathrm{imp}}_t \mathrm{IME}_{k,t,n} \\
& + \sum_b \sum_j \sum_t \widetilde{\mathrm{icm}}_{b,j,t} \mathrm{IM}_{b,j,t,n} + \sum_j \sum_s \sum_t \widetilde{\mathrm{ice}}_{j,t} \mathrm{IE}_{j,s,t,n} - \sum_c \sum_j \sum_s \sum_t \widetilde{\mathrm{bp}}_{c,t,n} \mathrm{PB}_{c,j,s,t} \Bigg)
\end{aligned} \tag{3}$$

Constraints

Biomass availability constraint Constraint (4) shows that the biomass harvested from each supply zone in each planning period and under each scenario cannot be greater than the available biomass feedstock. Available biomass feedstock itself is the product of the yield of biomass of each zone and the allocated land area and maximum fraction of residues that is allowed to be removed from supply zones.

$$\mathrm{HM}_{b,i,t,n} \le \nu\widetilde{\mathrm{ym}}_{b,i,t,n} l_{i,b} \quad \forall i,b,t,n \tag{4}$$

Material balance constraints Constraint (5) states that the amount of biomass transported from each supply zone in each planning period and under each scenario is equal to non-lost portion of harvested biomass.

$$\sum_{j}\sum_{m} \mathrm{TM}_{b,i,j,m,t,n} = \mathrm{HM}_{b,i,t,n}(1-\omega_{b,i,t}) \quad \forall i,b,t,n \tag{5}$$

Constraint (6) shows that under each scenario, the amount of transported biomass to each biorefinery in each planning period plus the biomass inventory level from the previous planning period considering biomass deterioration during storage equals the amount of consumed biomass feedstock at that biorefinery plus biomass inventory of the current planning period. Also, Constraint (7) indicates that under each scenario and for each biorefinery, sum of the produced amount of bioethanol and the inventory from the previous planning period is equal to the sum of the amount of bioethanol transported to demand zones and the inventory level of the current planning period.

$$\sum_{i}\sum_{m} \mathrm{TM}_{b,i,j,m,t,n} + (1-\theta_{b,t})\mathrm{IM}_{b,j,t-1,n} = \sum_{s} \mathrm{CM}_{b,j,s,t,n} + \mathrm{IM}_{b,j,t,n} \quad \forall b,j,t,n \tag{6}$$

$$\mathrm{PE}_{j,s,t} + \mathrm{IE}_{j,s,t-1,n} = \sum_{k}\sum_{m} \mathrm{TE}_{j,s,k,m,t,n} + \mathrm{IE}_{j,s,t,n} \quad \forall j,s,t,n \tag{7}$$

Conversion constraints Constraints (8) and (9) state that all types of biomass consumed at each biorefinery is converted into bioethanol and by-products according to their corresponding conversion factors through each technology, in each planning period, and under each scenario.

$$\mathrm{PE}_{j,s,t,n} = \sum_{b} \varphi_{b,s}\mathrm{CM}_{b,j,s,t,n} \quad \forall j,s,t,n \tag{8}$$

$$\mathrm{PB}_{c,j,s,t,n} = \sum_{b} \lambda_{b,s,c}\mathrm{CM}_{b,j,s,t,n} \quad \forall j,s,c,t,n \tag{9}$$

Capacity constraints Constraint (10) indicates that if a biorefinery is established in a location, its processing capacity should be within acceptable range, i.e., between minimum and maximum determined capacity levels.

$$\mathrm{lr}_s Y_{j,s,t} \le \mathrm{CR}_{j,s,t} \le \mathrm{ur}_s Y_{j,s,t} \quad \forall j,s,t \tag{10}$$

Constraint (11) limits the amount of biomass processed at each biorefinery, in each planning period, and under each scenario between minimum acceptable operating capacity of the biorefinery and its processing capacity.

$$\psi \mathrm{CR}_{j,s,t} \le \sum_{b} \mathrm{CM}_{b,j,s,t,n} \le \mathrm{CR}_{j,s,t} \quad \forall j,s,t,n \tag{11}$$

Constraint (12) ensures that at most only one biorefinery can be assigned to each candidate location.

$$\sum_{s} Y_{j,s,t} \le 1 \quad \forall j,t \tag{12}$$

Constraint (13) guarantees that once a biorefinery is established at a location it could not be shut down.

$$Y_{j,s,t-1} \le Y_{j,s,t} \quad \forall j,s,t \tag{13}$$

Constraint (14) shows that the processing capacity of a biorefinery can be increased but not decreased in subsequent planning periods.

$$\mathrm{CR}_{j,s,t-1} \le \mathrm{CR}_{j,s,t} \quad \forall j,s,t \tag{14}$$

Demand satisfaction constraint Constraint (15) states that total amount of bioethanol transported and imported to each demand zone should satisfy bioethanol demand of that demand zone, which itself is calculated through multiplying gasoline production of that demand zone by the planned blending fraction of bioethanol in gasoline.

$$\sum_{j}\sum_{s}\sum_{m} \mathrm{TE}_{j,s,k,m,t,n} + \mathrm{IME}_{k,t,n} \ge \mathrm{bf}_t\widetilde{\mathrm{pg}}_{k,t} \quad \forall k,t,n \tag{15}$$

Import constraint Constraint (16) limits the amount of bioethanol imported during each planning period and under each scenario to a predetermined upper bound.

$$\sum_{k} \mathrm{IME}_{k,t,n} \le \mathrm{uim}_t \quad \forall t,n \tag{16}$$

GHG emission constraint Constraint (17) restricts the amount of GHG emitted through biomass harvest, biomass transportation, biomass conversion to bioethanol and bioethanol transportation to a determined upper bound in each scenario.

$$\sum_i \sum_b \sum_t \mathrm{chm}_b \mathrm{HM}_{b,i,t,n} + \sum_b \sum_i \sum_j \sum_m \sum_t \mathrm{ctm}_{b,m} \mathrm{TM}_{b,i,j,m,t,n} \mathrm{dissr}_{i,j,m} + \sum_b \sum_j \sum_s \sum_t \mathrm{cpm}_{b,s} \mathrm{CM}_{b,j,s,t,n} + \sum_j \sum_s \sum_k \sum_m \sum_t \mathrm{cte}_m \mathrm{TE}_{j,s,k,m,t,n} \mathrm{disrd}_{j,k,m} \le \mathrm{uc} \quad \forall n \tag{17}$$

Decision variable constraints Constraints (18) and (19) specify binary and nonnegativity decision variables as:

$$\mathrm{CM}_{b,j,s,t,n}, \mathrm{CR}_{j,s,t}, \mathrm{HM}_{b,i,t,n}, \mathrm{IME}_{k,t,n}, \mathrm{PB}_{c,j,s,t,n}, \mathrm{PE}_{j,s,t,n}, \mathrm{TE}_{j,s,k,m,t,n}, \mathrm{TM}_{b,i,j,m,t,n} \ge 0 \quad \forall b,c,i,j,k,m,s,t,n \tag{18}$$

$$Y_{j,s,t} \in \{0,1\} \quad \forall j,s,t \tag{19}$$

Proposed robust stochastic-possibilistic programming (RSPP) approach

In order to handle the epistemic uncertainties and disruptions of the LBSC model under study, RSPP approach, proposed by Dehghan et al. (2018), is applied. In addition, inspired by the work of Pishvaee et al. (2012a), hard worst-case robust stochastic-possibilistic programming (HWRSPP) approach is proposed. To do so, in this section first possibilistic chance-constrained programming (with emphasis on Me measure), robust stochastic programming (RSP), robust possibilistic programming (RPP) and robust stochastic-possibilistic programming (RSPP) approaches will be explained briefly, and then, the proposed RSPP and HWRSPP models will be described.

Possibilistic chance-constrained programming (PCCP)

In possibilistic chance-constrained programming, each chance constraint should be satisfied with a determined confidence level (α) (Pishvaee et al. 2012b). Prade and Dubois (1988) introduced two basic fuzzy measures, possibility (Pos) and necessity (Nec) measures, to help include such a level of confidence in PCCP models. Pos and Nec measures have extreme optimistic and pessimistic attitudes toward chance constraints, respectively. In order to moderate such extreme attitudes, Liu and Liu (2002) introduced credibility (Cr) measure as the average of Pos and Nec measures. Recently, Xu and Zhou (2013) have introduced Me measure which is a weighted average of Pos and Nec measures and is an expanded form of the Cr measure. Me measure is defined by including a parameter named optimistic–pessimistic parameter, which is actually the weight of Pos measure in the definition of Me measure.

$$\begin{aligned} \mathrm{Me}\{A\} &= \gamma.\mathrm{Pos}\{A\} + (1-\gamma).\mathrm{Nec}\{A\} \\ &= \mathrm{Nec}\{A\} + \gamma(\mathrm{Pos}\{A\} - \mathrm{Nec}\{A\}) \end{aligned} \tag{20}$$

Here, γ ($0 \le \gamma \le 1$) is the optimistic–pessimistic parameter. It is worthwhile to note that Me measure converts to Nec, Pos and Cr measures when γ takes values of 0, 1 and 0.5.

Consider a trapezoidal fuzzy variable $\tilde{\xi} = (r_1, r_2, r_3, r_4)$, $r_1 \le r_2 \le r_3 \le r_4$ which its membership function is as Eq. (21):

$$\mu_{(x)} = \begin{cases} \frac{x-r_1}{r_2-r_1} & r_1 \le x \le r_2 \\ 1 & r_2 \le x \le r_3 \\ \frac{r_4-x}{r_4-r_3} & r_3 \le x \le r_4 \\ 0 & \text{otherwise} \end{cases} \tag{21}$$

the fuzzy measures of the events $\tilde{\xi} \le x$ and $\tilde{\xi} \ge x$ are presented as Eqs. (22) and (23) (Xu and Zhou 2013).

$$\mathrm{Me}\left\{\tilde{\xi} \le x\right\} = \begin{cases} 0 & x \le r_1 \\ \gamma \frac{x-r_1}{r_2-r_1} & r_1 \le x \le r_2 \\ \gamma & r_2 \le x \le r_3 \\ \gamma + (1-\gamma)\frac{x-r_3}{r_4-r_3} & r_3 \le x \le r_4 \\ 1 & x \ge r_4 \end{cases} \tag{22}$$

$$\mathrm{Me}\left\{\tilde{\xi} \ge x\right\} = \begin{cases} 1 & x \le r_1 \\ \gamma + (1-\gamma)\frac{r_2-x}{r_2-r_1} & r_1 \le x \le r_2 \\ \gamma & r_2 \le x \le r_3 \\ \gamma \frac{r_4-x}{r_4-r_3} & r_3 \le x \le r_4 \\ 0 & x \ge r_4 \end{cases} \tag{23}$$

Equation (24) shows the expected value of the fuzzy variable $\tilde{\xi}$ based on *Me* measure (Xu and Zhou 2013).

$$E^{\mathrm{Me}}[\xi] = \int_0^{+\infty} \mathrm{Me}\{\xi \ge x\}\mathrm{d}x - \int_{-\infty}^{0} \mathrm{Me}\{\xi \le x\}\mathrm{d}x \tag{24}$$

Based on Me measure, Xu and Zhou (2013) present the expected value of the fuzzy trapezoidal variable $\tilde{\xi}$ as Eq. (25):

$$E^{\mathrm{Me}}[\xi] = \begin{cases} \frac{\gamma}{2}(r_1+r_2) + \left(\frac{1-\gamma}{2}\right)(r_3+r_4) & r_4 \le 0 \\ \frac{\gamma}{2}(r_1+r_2) + \frac{\gamma r_4^2 - (1-\gamma) r_3^2}{2(r_4-r_3)} & r_3 \le 0 \le r_4 \\ \frac{\gamma}{2}(r_1+r_2+r_3+r_4) & r_2 \le 0 \le r_3 \\ \frac{(1-\gamma) r_2^2 - \gamma r_1^2}{2(r_4-r_3)} + \frac{\gamma}{2}(r_3+r_4) & r_1 \le 0 \le r_2 \\ \left(\frac{1-\gamma}{2}\right)(r_1+r_2) + \frac{\gamma}{2}(r_3+r_4) & 0 \le r_1 \end{cases} \tag{25}$$

According to Mousazadeh et al. (2018), the crisp counterparts of $\mathrm{Me}\left\{\tilde{\xi} \le x\right\} \ge \alpha$ and $\mathrm{Me}\left\{\tilde{\xi} \ge x\right\} \ge \alpha$, for all confidence levels equal to or greater than 0.5 ($\alpha \ge 0.5$) and $0 \le \gamma \le 0.5$, would be as Eqs. (26) and (27):

$$\mathrm{Me}\left\{\tilde{\xi} \le x\right\} \ge \alpha \Leftrightarrow \gamma + (1-\gamma)\frac{x - r_3}{r_4 - r_3} \ge \alpha \Leftrightarrow x \ge \frac{(\alpha-\gamma)r_4 + (1-\alpha)r_3}{(1-\gamma)} \tag{26}$$

$$\mathrm{Me}\left\{\tilde{\xi} \ge x\right\} \ge \alpha \Leftrightarrow \gamma + (1-\gamma)\frac{r_2 - x}{r_2 - r_1} \ge \alpha \Leftrightarrow x \le \frac{(\alpha-\gamma)r_1 + (1-\alpha)r_2}{(1-\gamma)} \tag{27}$$

For $\gamma \ge 0.5$, interested reader can see Mousazadeh et al. (2018).

Robust stochastic programming (RSP) approach

The framework of robust scenario-based stochastic programming approach, developed by Mulvey et al. (1995), consists of two types of robustness: 'solution robustness,' which states that the solution should remain nearly optimal in all scenarios, and 'model robustness,' which states that the solution should remain nearly feasible in all scenarios. The objective function has penalty functions for both model and solution robustness, weighted by parameters in order to capture DMs' preference between the two. For describing RSP approach, consider the following liner programming (LP) model (28):

$$\begin{aligned} &\min z = c_n \cdot X_n + d_n \cdot Y \\ &\text{s.t.} \\ &A_n \cdot X_n + B_n \cdot Y \ge e_n \\ &X_n, Y \ge 0 \end{aligned} \tag{28}$$

where X_n and Y denote the vector of decision variables that should be determined under the uncertainty of model parameters. While Y is the vector of first stage decision variables, X_n is the vector of second stage decision variables. Also, c_n, d_n and e_n are vectors of uncertain parameters and A_n and B_n represent coefficient matrices of uncertain parameters in the model constraints. Parameter n denotes each scenario, $n \in S$, and S is a finite set of scenarios, $S = \{1, 2, \ldots N\}$. According to Mulvey et al. (1995), the robust stochastic programming (RSP) model could be written as model (29):

$$\begin{aligned} &\min E[z] + \sigma \sum_n P_n \left(c_n \cdot X_n + d_n \cdot Y - E[z]\right)^2 + \omega \sum_n P_n S_n \\ &\text{s.t.} \\ &A_n \cdot X_n + B_n \cdot Y + S_n \ge e_n \\ &X_n, Y, S_n \ge 0 \end{aligned} \tag{29}$$

Here, the expected value of the objective function ($E[z]$) could be stated as Eq. (30):

$$E[z] = \sum_n P_n \left(c_n \cdot X_n + d_n \cdot Y\right) \tag{30}$$

While the first term in the objective function minimizes the expected value of z, the second term tries to provide solution robustness by minimizing the solution variance under all scenarios. And the third term intends to minimize violations from constraints by penalizing such violations (S_n). The weights σ and ω denote the importance weight of solution and model robustness over other terms in the objective function. Yu and Li (2000) replaced the quadratic term in the objective function with an absolute deviation term to reduce the computational effort required and proposed model (31):

$$\begin{aligned} &\min = E[z] + \sigma \sum_n P_n \left|c_n \cdot X_n + d_n \cdot Y - E[z]\right| + \omega \sum_n P_n S_n \\ &\text{s.t.} \\ &A_n \cdot X_n + B_n \cdot Y + S_n \ge e_n \\ &X_n, Y, S_n \ge 0 \end{aligned} \tag{31}$$

Then, they linearized the absolute term in the objective function by introducing a nonnegative variable (ζ_n) and an additional constraint as in model (32):

$$\begin{aligned} &\min = E[z] + \sigma \sum_n P_n \left(c \cdot X_n + d \cdot Y - E[z] + 2\zeta_n\right) + \omega \sum_n P_n S_n \\ &\text{s.t.} \\ &A_n \cdot X_n + B_n \cdot Y + S_n \ge e_n \\ &c \cdot X_n + d \cdot Y - E[z] + \zeta_n \ge 0 \\ &X_n, Y, S_n, \ \zeta_n \ge 0 \end{aligned} \tag{32}$$

Robust possibilistic programming (RPP) approach

The concept of RPP approach was first introduced by Pishvaee et al. (2012a), which is similar to RSP approach. To describe RPP approach consider the following LP model (33):

$$\begin{aligned}&\min z = \widetilde{c}\cdot X + \widetilde{d}\cdot Y\\&\text{s.t.}\\&A\cdot X + B\cdot Y \le \widetilde{b}\\&C\cdot X + D\cdot Y \ge \widetilde{e}\\&X, Y \ge 0\end{aligned} \tag{33}$$

Here, X and Y are the vector of decision variables that should be determined under the epistemic uncertainty (fuzziness) of model parameters. And $\widetilde{c}$, $\widetilde{d}$, $\widetilde{b}$, and $\widetilde{e}$ denote vectors of parameters with epistemic uncertainty. Also A, B, C and D represent coefficient matrices of model constraints. Regarding Pishvaee et al. (2012a) and applying *Me* measure the basic possibilistic programming (BPP) model could be formulated as model (34):

$$\begin{aligned}&\min E[z] = E\left[\widetilde{c}\right]\cdot X + E\left[\widetilde{d}\right]\cdot Y\\&\text{s.t.}\\&\text{Me}\left\{A\cdot X + B\cdot Y \le \widetilde{b}\right\} \ge \alpha\\&\text{Me}\left\{C\cdot X + D\cdot Y \ge \widetilde{e}\right\} \ge \beta\\&X, Y \ge 0\end{aligned} \tag{34}$$

where α and β are the minimum confidence levels of chance constraints, determined by DMs. Assuming that the imprecise parameters have trapezoidal possibility distributions and applying Eq. (25) (for $c_{(1)}$ and $d_{(1)} \ge 0$), and Eqs. (26) and (27) (for $\gamma < 0.5$), the crisp counterpart of the model could be written as model (35):

$$\begin{aligned}&\min E[z] = \left[\frac{1-\gamma}{2}\left(c_{(1)}+c_{(2)}\right)+\frac{\gamma}{2}\left(c_{(3)}+c_{(4)}\right)\right]\cdot X\\&\quad+\left[\frac{1-\gamma}{2}\left(d_{(1)}+d_{(2)}\right)+\frac{\gamma}{2}\left(d_{(3)}+d_{(4)}\right)\right]\cdot Y\\&\text{s.t.}\\&A\cdot X + B\cdot Y \le \left[\frac{(\alpha-\gamma)b_{(1)}+(1-\alpha)b_{(2)}}{1-\gamma}\right]\\&C\cdot X + D\cdot Y \ge \left[\frac{(\beta-\gamma)e_{(4)}+(1-\beta)e_{(3)}}{1-\gamma}\right]\\&X, Y \ge 0, \quad 0.5 \le \alpha, \beta \le 1\end{aligned} \tag{35}$$

Pishvaee et al. (2012a) proposed RPP approach by adding some terms to the objective function to provide optimality and feasibility robustness. Feasibility robustness states that the solution should remain feasible for (almost) all possible values of uncertain parameters and optimality robustness states that the solution should approximately remain optimal or have minimum deviation from the optimal value for (almost) all possible values of uncertain parameters (Pishvaee et al. 2012a). The RPP model can be written as model (36):

$$\begin{aligned}&\min E[z] + \phi\left(z_{\max} - z_{\min}\right) + \delta_1\left[\frac{(\alpha-\gamma)b_{(1)}+(1-\alpha)b_{(2)}}{1-\gamma} - b_{(1)}\right]\\&\quad+\delta_2\left[e_{(4)} - \frac{(\beta-\gamma)e_{(4)}+(1-\beta)e_{(3)}}{1-\gamma}\right]\\&\text{s.t.}\\&A\cdot X + B\cdot Y \le \left[\frac{(\alpha-\gamma)b_{(1)}+(1-\alpha)b_{(2)}}{1-\gamma}\right]\\&C\cdot X + D\cdot Y \ge \left[\frac{(\beta-\gamma)e_{(4)}+(1-\beta)e_{(3)}}{1-\gamma}\right]\\&X, Y \ge 0, \quad 0.5 \le \alpha, \beta \le 1\end{aligned} \tag{36}$$

The first term in the objective function minimizes the expected value of z, whereas the second term provides optimality robustness by minimizing the difference between the two extreme possible values of z, in which $z_{\max}$ and $z_{\min}$ can be defined as (37):

$$\begin{aligned}z_{\max} &= c_{(4)}\cdot X + d_{(4)}\cdot Y\\z_{\min} &= c_{(1)}\cdot X + d_{(1)}\cdot Y\end{aligned} \tag{37}$$

The third and fourth terms provide feasibility robustness by penalizing violations from the right-hand side of chance constraints. Indeed, the amount of violation is the difference between right-hand side of chance constraints and their worst-case values, i.e., $b_{(1)}$ and $e_{(4)}$. Parameters δ_1 and δ_2 indicate unit penalty cost of possible violations from right-hand side of the chance constraints. And parameter ϕ denotes the importance weight of the term optimality robustness against other terms of the objective function.

Robust stochastic-possibilistic programming (RSPP) approach

Integrating RSP and RPP approaches, Dehghan et al. (2018) proposed RSPP approach to make models robust against simultaneous randomness and epistemic uncertainty of parameters. In order to briefly describe RSPP approach consider the LP model (38):

$$\begin{aligned}&\min z = \widetilde{c}_n\cdot X_n + \widetilde{d}_n\cdot Y\\&\text{s.t.}\\&A\cdot X_n + B\cdot Y \le \widetilde{b}_n\\&C\cdot X_n + D\cdot Y \ge \widetilde{e}\\&X_n, Y \ge 0\end{aligned} \tag{38}$$

where Y is the vector of first stage decision variables and X_n is the vector of second stage decision variables. Parameters A, B, C and D represent coefficient matrices of model constraints. While $\widetilde{e}$ denotes vector of parameters with epistemic uncertainty, $\widetilde{c}_n$, $\widetilde{d}_n$ and $\widetilde{b}_n$ are vectors of parameters with randomly epistemic uncertainty (assuming that a given parameter $\widetilde{\xi}_n$ has trapezoidal possibility distribution, $\widetilde{\xi}_n = \left(r_{(1n)}, r_{(2n)}, r_{(3n)}, r_{(4n)}\right),\ r_{(1n)} \le r_{(2n)} \le r_{(3n)} \le r_{(4n)}\ \forall n$) Parameter n represents each scenario $n \in S$ and S is a finite set of scenarios $S = \{1, 2, \ldots N\}$. Applying the *Me* measure, basic stochastic-possibilistic programming (BSPP) model can be formulated as model (39):

$$\begin{aligned} &\min E[z] = E\left[\widetilde{c}_n\right] \cdot X + E\left[\widetilde{d}_n\right] \cdot Y \\ &\text{s.t.} \\ &\mathrm{Me}\left\{A \cdot X_n + B \cdot Y \le \widetilde{b}_n\right\} \ge \alpha_n \\ &\mathrm{Me}\left\{C \cdot X_n + D \cdot Y \ge \widetilde{e}\right\} \ge \beta_n \\ &X_n, Y \ge 0 \end{aligned} \tag{39}$$

where α_n and β_n are the minimum confidence levels of chance constraints under each scenario. Assuming that the imprecise parameters have trapezoidal possibility distributions and applying Eqs. (25), (for $c_{(1n)}$ and $d_{(1n)} \ge 0$), (26) and (27) (for $\gamma < 0.5$), the crisp counterpart of the model could be stated as model (40):

$$\begin{aligned} &\min E[z] = \sum_n P_n \left(\left[\frac{1-\gamma}{2}\left(c_{(1n)} + c_{(2n)}\right) + \frac{\gamma}{2}\left(c_{(3n)} + c_{(4n)}\right) \right] \cdot X_n + \left[\frac{1-\gamma}{2}\left(d_{(1n)} + d_{(2n)}\right) + \frac{\gamma}{2}\left(d_{(3n)} + d_{(4n)}\right) \right] \cdot Y \right) \\ &\text{s.t.} \\ &A \cdot X_n + B \cdot Y \le \left[\frac{(\alpha_n - \gamma) b_{(1n)} + (1 - \alpha_n) b_{(2n)}}{1-\gamma} \right] \\ &C \cdot X_n + D \cdot Y \ge \left[\frac{(\beta_n - \gamma) e_{(4)} + (1 - \beta_n) e_{(3)}}{1-\gamma} \right] \\ &X_n, Y \ge 0, \quad 0.5 \le \alpha_n, \beta_n \le 1 \end{aligned} \tag{40}$$

In order to provide optimality and feasibility robustness against parameters with randomly epistemic uncertainty, Dehghan et al. (2018) proposed RSPP approach. This approach takes advantage of RSP and RPP approaches. Based on *Me* measure, the RSPP model could be written as model (41):

$$\begin{aligned} &\min E[z] + \phi\left(z_{\max} - z_{\min}\right) + \sigma \sum_n P_n \left| E[z] - E\left[z_n\right] \right| \\ &+ \delta_1 \sum_n P_n \left[\frac{(\alpha_n - \gamma) b_{(1n)} + (1 - \alpha_n) b_{(2n)}}{1-\gamma} - b_{(1n)} \right] \\ &+ \delta_2 \left[e_{(4)} - \frac{(\beta_n - \gamma) e_{(4)} + (1 - \beta_n) e_{(3)}}{1-\gamma} \right] \\ &\text{s.t.} \\ &A \cdot X_n + B \cdot Y \le \left[\frac{(\alpha_n - \gamma) b_{(1n)} + (1 - \alpha_n) b_{(2n)}}{1-\gamma} \right] \\ &C \cdot X_n + D \cdot Y \ge \left[\frac{(\beta_n - \gamma) e_{(4)} + (1 - \beta_n) e_{(3)}}{1-\gamma} \right] \\ &X_n, Y \ge 0, \quad 0.5 \le \alpha_n, \beta_n \le 1 \end{aligned} \tag{41}$$

Like RPP model presented in model (36), the second term provides optimality robustness of the solution under fuzziness of parameters by minimizing the difference between the two extreme possible values of z which is called 'possibilistic deviation.' Maximum and minimum possible values of z ($z_{\max}$ and $z_{\min}$) could be obtained as (42):

$$\begin{aligned} z_{\max} &= \sum_n P_n c_{(4n)} \cdot X_n + \sum_n P_n d_{(4n)} \cdot Y \\ z_{\min} &= \sum_n P_n c_{(1n)} \cdot X_n + \sum_n P_n d_{(1n)} \cdot Y \end{aligned} \tag{42}$$

Like RSP approach presented in model (31), the third term provides optimality robustness of the solution under randomness of parameters by minimizing 'stochastic deviation' which is the deviation of the expected value of the objective function under different scenarios ($E[z_n]$) over and under the expected value of z. The expected value of the objective function under each scenario ($E[z_n]$) is stated as Eq. (43):

$$E\left[z_n\right] = \left[\frac{1-\gamma}{2}\left(c_{(1n)} + c_{(2n)}\right) + \frac{\gamma}{2}\left(c_{(3n)} + c_{(4n)}\right) \right] \cdot X_n + \left[\frac{1-\gamma}{2}\left(d_{(1n)} + d_{(2n)}\right) + \frac{\gamma}{2}\left(d_{(3n)} + d_{(4n)}\right) \right] \cdot Y \tag{43}$$

The same as RPP model, the last two terms in the objective function account for feasibility robustness of the solution. Parameters δ_1 and δ_2 denote unit penalty cost of violations from right-hand side of the chance constraints. Also parameters ϕ and σ are the importance weights of optimality robustness under fuzziness and randomness of parameters against the other terms of the objective function. The absolute term in the objective function can be linearized in the way described in model (32). Thus, the linearized form of the RSPP model is as model (44):

$$\begin{aligned}
&\min E[z] + \phi\left(z_{\max} - z_{\min}\right) + \sigma \sum_n P_n\left(E[z] - E\left[z_n\right] + 2\zeta_n\right) \\
&+ \delta_1 \sum_n P_n \left[\frac{(\alpha_n - \gamma) b_{(1n)} + (1-\alpha_n) b_{(2n)}}{1-\gamma} - b_{(1n)}\right] \\
&+ \delta_2 \left[e_{(4)} - \frac{(\beta_n - \gamma) e_{(4)} + (1-\beta_n) e_{(3)}}{1-\gamma}\right] \\
&\text{s.t.} \\
&A \cdot X_n + B \cdot Y \le \left[\frac{(\alpha_n - \gamma) b_{(1n)} + (1-\alpha_n) b_{(2n)}}{1-\gamma}\right] \\
&C \cdot X_n + D \cdot Y \ge \left[\frac{(\beta_n - \gamma) e_{(4)} + (1-\beta_n) e_{(3)}}{1-\gamma}\right] \\
&E[z] - E\left[z_n\right] + \zeta_n \ge 0 \\
&X_n, Y, \zeta_n \ge 0, \quad 0.5 \le \alpha_n, \beta_n \le 1
\end{aligned} \tag{44}$$

The proposed basic stochastic-possibilistic programming (BSPP) model

In order to avoid prolongation, the proposed LBSC model described in Sect. 3.2 (Eqs. 3-19) can be rewritten in a compact form as model (45):

$$\begin{aligned}
&\min Z = \widetilde{f} \cdot Y + \widetilde{g} \cdot X + \widetilde{h} \cdot X_n - \widetilde{l} \cdot X_n \\
&\text{s.t.} \\
&X_n \le A \cdot \widetilde{\mathrm{ym}}_n \\
&B \cdot X_n = 0 \\
&C \cdot Y \le X \le D \cdot Y \\
&E \cdot X_n \le X \\
&F \cdot Y \ge 0 \\
&G \cdot Y \le 1 \\
&X_n \ge H \cdot \widetilde{\mathrm{pg}} \\
&K \cdot X_n \le \mathrm{U} \\
&Y \in \{0,1\}, X, X_n \ge 0
\end{aligned} \tag{45}$$

where vectors f and g represent fixed and variable setup costs of biorefineries, respectively, vector h indicates other costs and vector l corresponds to sale price of by-products. Matrices A, B, C, D, E, F, G, H, K and U denote coefficient matrices of the model constraints. And parameters ym_n and pg represent yields of biomass and gasoline production of oil refineries. Vector Y represents binary variables, vector X indicates capacity of biorefineries, and vector X_n displays other variables.

Assuming that imprecise parameters have trapezoidal possibility distributions and applying *Me* measure, the BSPP model of the compact model (45) can be written as (46):

$$\begin{aligned}
&\min E[Z] = E\left[\widetilde{f}\right] \cdot Y + E\left[\widetilde{g}\right] \cdot X + E\left[\widetilde{h}\right] \cdot X_n - E\left[\widetilde{l}\right] \cdot X_n \\
&\text{s.t.} \\
&\mathrm{Me}\left\{X_n \le A \cdot \widetilde{\mathrm{ym}}\right\} \ge \alpha_n \\
&B \cdot X_n = 0 \\
&C \cdot Y \le X \le D \cdot Y \\
&E \cdot X_n \le X \\
&F \cdot Y \ge 0 \\
&G \cdot Y \le 1 \\
&\mathrm{Me}\left\{X_n \ge H \cdot \widetilde{\mathrm{pg}}\right\} \ge \beta_n \\
&K \cdot X_n \le U \\
&Y \in \{0,1\}, X, X_n \ge 0
\end{aligned} \tag{46}$$

where α_n and β_n are the minimum confidence levels of chance constraints under each scenario. It is assumed that α_n and β_n are greater than 0.5.

Based on model (40) as previously described, the crisp counterpart of the BSPP model could be presented as model (47):

$$\begin{aligned}
&\min E[Z] = \left[\frac{1-\gamma}{2}\left(f_{(1)} + f_{(2)}\right) + \frac{\gamma}{2}\left(f_{(3)} + f_{(4)}\right)\right] \cdot Y \\
&+ \left[\frac{1-\gamma}{2}\left(g_{(1)} + g_{(2)}\right) + \frac{\gamma}{2}\left(g_{(3)} + g_{(4)}\right)\right] \cdot X \\
&+ \sum_n p_n \left[\frac{1-\gamma}{2}\left(h_{(1)} + h_{(2)}\right) + \frac{\gamma}{2}\left(h_{(3)} + h_{(4)}\right)\right] \cdot X_n \\
&- \sum_n p_n \left[\frac{1-\gamma}{2}\left(l_{(1)} + l_{(2)}\right) + \frac{\gamma}{2}\left(l_{(3)} + l_{(4)}\right)\right] \cdot X_n \\
&\text{s.t.} \\
&X_n \le A \cdot \left[\frac{(\alpha_n - \gamma)\mathrm{ym}_{(1n)} + (1-\alpha_n)\mathrm{ym}_{(2n)}}{1-\gamma}\right] \\
&B \cdot X_n = 0 \\
&C \cdot Y \le X \le D \cdot Y \\
&E \cdot X_n \le X \\
&F \cdot Y \ge 0 \\
&G \cdot Y \le 1 \\
&X_n \ge H \cdot \left[\frac{(\beta_n - \gamma)\mathrm{pg}_{(4)} + (1-\beta_n)\mathrm{pg}_{(3)}}{1-\gamma}\right] \\
&K \cdot X_n \le U \\
&Y \in \{0,1\}, X, X_n \ge 0, \quad 0.5 \le \alpha_n, \beta_n \le 1
\end{aligned} \tag{47}$$

The proposed robust stochastic-possibilistic programming (RSPP) model

Since the proposed BSPP model is sensitive neither to the deviation of the objective function from its expected value nor to the violations from the right-hand sides of chance constraints, RSPP model is applied in this section to overcome this disadvantage of the BSPP model. Regarding model (41) as previously explained the proposed RSPP model of the compact model (45) could be presented as model (48):

$$\begin{aligned}
&\min E[Z] + \phi\left(Z_{\max} - Z_{\min}\right) + \sigma \sum_n p_n \left| E[Z] - E\left[Z_n\right] \right| \\
&\quad + \delta_1 \sum_n p_n A \cdot \left[\frac{(\alpha_n - \gamma)\mathrm{ym}_{(1n)} + (1-\alpha_n)\mathrm{ym}_{(2n)}}{1-\gamma} - \mathrm{ym}_{(1n)} \right] \\
&\quad + \delta_2 H \cdot \left[\mathrm{pg}_{(4)} - \frac{(\beta_n - \gamma)\mathrm{pg}_{(4)} + (1-\beta_n)\mathrm{pg}_{(3)}}{1-\gamma} \right] \\
&\text{s.t.} \\
&X_n \le A \cdot \left[\frac{(\alpha_n - \gamma)\mathrm{ym}_{(1n)} + (1-\alpha_n)\mathrm{ym}_{(2n)}}{1-\gamma} \right] \\
&B \cdot X_n = 0 \\
&C \cdot Y \le X \le D \cdot Y \\
&E \cdot X_n \le X \\
&F \cdot Y \ge 0 \\
&G \cdot Y \le 1 \\
&X_n \ge H \cdot \left[\frac{(\beta_n - \gamma)\mathrm{pg}_{(4)} + (1-\beta_n)\mathrm{pg}_{(3)}}{1-\gamma} \right] \\
&K \cdot X_n \le U \\
&Y \in \{0,1\}, X, X_n \ge 0, \quad 0.5 \le \alpha_n, \beta_n \le 1
\end{aligned} \tag{48}$$

While the first term in the objective function minimizes the expected value of z, the second term tries to minimize possibilistic deviation in order to control the optimality robustness under fuzziness of parameters. Regarding Eq. (42), $z_{\max}$ and $z_{\min}$ could be presented as (49):

$$\begin{aligned}
Z_{\max} &= f_{(4)} \cdot Y + g_{(4)} \cdot X + \sum_n p_n h_{(4)} \cdot X_n - \sum_n p_n l_{(1)} \cdot X_n \\
Z_{\min} &= f_{(1)} \cdot Y + g_{(1)} \cdot X + \sum_n p_n h_{(1)} \cdot X_n - \sum_n p_n l_{(4)} \cdot X_n
\end{aligned} \tag{49}$$

The third term tries to minimize stochastic deviation in order to control the optimality robustness under randomness of parameters. Based on Eq. (43), $E[z_n]$ could be defined as Eq. (50):

$$\begin{aligned}
E\left[Z_n\right] = &\left[\frac{1-\gamma}{2}\left(f_{(1)} + f_{(2)}\right) + \frac{\gamma}{2}\left(f_{(3)} + f_{(4)}\right) \right] \cdot Y + \left[\frac{1-\gamma}{2}\left(g_{(1)} + g_{(2)}\right) + \frac{\gamma}{2}\left(g_{(3)} + g_{(4)}\right) \right] \cdot X \\
&+ \left[\frac{1-\gamma}{2}\left(h_{(1)} + h_{(2)}\right) + \frac{\gamma}{2}\left(h_{(3)} + h_{(4)}\right) \right] \cdot X_n - \left[\frac{1-\gamma}{2}\left(l_{(1)} + l_{(2)}\right) + \frac{\gamma}{2}\left(l_{(3)} + l_{(4)}\right) \right] \cdot X_n
\end{aligned} \tag{50}$$

The last two terms have been added to the objective function to control the feasibility robustness of the solution. This is done by minimizing the difference between right-hand side of chance constraints and their worst-case values, i.e., $\mathrm{ym}_{(1)}$and $\mathrm{pg}_{(4)}$. Parameters δ_1 and δ_2 denote unit penalty costs of violations from right-hand side of chance constraints. Also parameters ϕ and σ specify the importance weight of optimality robustness under fuzziness and randomness of parameters against the other terms of the objective function. As described previously in model (44), the third term of the objective function could be linearized by adding an additional variable (ζ_n) and a constraint to the model as presented in model (51).

$$\begin{aligned}
&\min E[Z] + \phi\left(Z_{\max} - Z_{\min}\right) + \sigma \sum_n p_n \left\{ E[Z] - E\left[Z_n\right] + 2\zeta_n \right\} \\
&\quad + \delta_1 \sum_n p_n A \cdot \left[\frac{(\alpha_n - \gamma)\mathrm{ym}_{(1n)} + (1-\alpha_n)\mathrm{ym}_{(2n)}}{1-\gamma} - \mathrm{ym}_{(1)} \right] \\
&\quad + \delta_2 H \cdot \left[\mathrm{pg}_{(4)} - \frac{(\beta_n - \gamma)\mathrm{pg}_{(4)} + (1-\beta_n)\mathrm{pg}_{(3)}}{1-\gamma} \right] \\
&\text{s.t.} \\
&X_n \le A \cdot \left[\frac{(\alpha_n - \gamma)\mathrm{ym}_{(1n)} + (1-\alpha_n)\mathrm{ym}_{(2n)}}{1-\gamma} \right] \\
&B \cdot X_n = 0 \\
&C \cdot Y \le X \le D \cdot Y \\
&E \cdot X_n \le X \\
&F \cdot Y \ge 0 \\
&G \cdot Y \le 1 \\
&X_n \ge H \cdot \left[\frac{(\beta_n - \gamma)\mathrm{pg}_{(4)} + (1-\beta_n)\mathrm{pg}_{(3)}}{1-\gamma} \right] \\
&K \cdot X_n \le U \\
&E[z] - E\left[z_n\right] + \zeta_n \ge 0 \\
&Y \in \{0,1\}, X, X_n, \zeta_n \ge 0, \quad 0.5 \le \alpha_n, \beta_n \le 1
\end{aligned} \tag{51}$$

The proposed hard worst-case robust stochastic-possibilistic programming (HWRSPP) approach

According to Pishvaee et al. (2012a), in the worst-case robust programming, the worst-case value of imprecise parameters in the objective function and constraints are considered. So, it is the most conservative and risk-averse

version of robust programming. Adopting the hard worst-case approach, HWRSPP model of the compact model (45) for trapezoidal form of fuzzy parameters can be formulated as model (52):

$$\begin{aligned}
&\min Z_{\max} = f_{(4)} \cdot Y + g_{(4)} \cdot X + \sum_n p_n h_{(4)} \cdot X_n - \sum_n p_n l_{(1)} \cdot X_n \\
&s.t. \\
&X_n \le A \cdot ym_{(1n)} \\
&B \cdot X_n = 0 \\
&C \cdot Y \le X \le D \cdot Y \\
&E \cdot X_n \le X \\
&F \cdot Y \ge 0 \\
&G \cdot Y \le 1 \\
&X_n \ge H \cdot \mathrm{pg}_{(4)} \\
&K \cdot X_n \le U \\
&Y \in \{0,1\}, X, X_n \ge 0
\end{aligned} \tag{52}$$

Case study

In order to assess the desirability and performance of the proposed models, a real case study in Iran is conducted. Iran is a large country with high climate variation in different regions and since cereals such as rice, wheat, barley and corn form a big part of the diet in Iran, these cereal crops are widely cultivated. Thus, in this study, rice straw, wheat straw, barley straw and corn stover are considered as lignocellulosic biomass feedstocks. And 23 provinces of Iran which are responsible for production of more than 90% of above-mentioned cereals are considered as biomass supply zones. Also, 14 provinces of Iran are chosen as potential locations for biorefineries. These locations are selected based on some social, geographical and underlying structural considerations.

Since, it is supposed that the produced bioethanol at biorefineries will be sent to oil refineries in order to be blended into gasoline, all nine oil refineries of Iran, namely Tabriz, Shiraz, Bandar Abbas, Isfahan, Kermanshah, Abadan, Arak, Tehran and Lavan, are considered to be bioethanol demand zones.

Thermochemical or biochemical technologies are considered to be employed in biorefineries to convert biomass into bioethanol and by-products. By-products of these two technologies are mixed alcohols and bioelectricity, respectively. Biomass and bioethanol can be transported by roadway or railway (if available) through the network.

Bioethanol demand of each demand zone (oil refinery) is obtained by multiplying the quantity of gasoline produced at each oil refinery by the blending fraction of bioethanol (constraint (15)). The blending fraction (bf_t) is supposed to be 5% for all time periods regarding "National Document for Knowledge-based Development of Renewable energies" (2015). And, Table 2 reports gasoline production rate of oil refineries at year 2014. The planning horizon is 1 year, which is broken into 4 three-month planning periods. First planning period is considered to be summer when biomass availability is at its maximum and leads to a less cost. Four drought disruption scenarios are defined including no drought, mild drought, moderate and severe drought with occurrence probabilities of 0.05, 0.15, 0.7 and 0.1, respectively. Yield of different biomass types of each supply zone in each planning period and under each scenario is presented by a fuzzy number. The loss in yield of biomass for different supply zones is calculated based on meteorological drought zoning of Iran ("Yearbook of the National Drought Warning and Monitoring Center (NDWMC)," 2017–2018). The most possible values of the fuzzy parameters, including biomass yield of "no drought scenario," gasoline production of each oil refinery and different costs and sale prices are defined based on historical data and the related literature. Other values of fuzzy parameters are estimated according to experts' opinion. Some other assumptions considered to conduct the case study are stated below:

Table 2 Gasoline production rate of oil refineries in year 2014 ("Iran Statistical Yearbook of oil products consumption" 2014)

No.	Oil refinery	Gasoline production rate (MLD[a])
1	Tabriz	3.2
2	Shiraz	1.8
3	Bandar Abbas	8.5
4	Isfahan	11.3
5	Kermanshah	0.6
6	Abadan	10
7	Arak	14.6
8	Tehran	6.6
9	Lavan	2.2

[a]Million liters per day

- Biomass supply zones and potential location of biorefineries are assumed to be in the capital city of the provinces.
- It is assumed that there is a demand for the total amount of by-products produced.
- Maximum fraction of residues that is allowed to be removed from supply zones (ν) is considered to be 0.3 (constraint (4) of the model).
- Loss fraction of harvesting biomass ($\omega_{b,i,t}$) is supposed to be 0.05 for all types of biomass, in all supply zones and

Table 3 The statistics of BSPP, RSPP and HWRSPP models

Model	No. of equations	No. of continuous variables	No. of binary variables	Average execution time of each run (s)
BSPP	7197	55,601	112	187
RSPP	7206	55,608	112	76
HWRSPP	7197	55,601	112	75

all planning periods (constraint (5)) according to (Čuček et al. 2014).

- Thermochemical and biochemical biorefineries with a processing capacity of less than 1 million tons per year (MTPY) are not economical, while with a processing capacity of more than 3 MTPY are not yet technically feasible (Osmani and Zhang 2017). Therefore, each biorefinery is considered to have a processing capacity between 250,000 and 750,000 tons per planning period (constraint (10)).
- Minimum acceptable efficiency percentage for biorefinery capacity (ψ) is assumed to be 0.8 (constraint (11)).
- According to policymakers, maximum amount of bioethanol allowed to be imported is considered to be less than 10 million gallons per period (constraint (16)).

Required input data of the case study and the related references are provided through Tables 15, 16, 17, 18, 19, 20, 21, 22 and 23 of Appendix: key deterministic parameters and nominal values of uncertain parameters (Tables 15 and 16), allocated land area of supply zones for cultivation of different biomass feedstocks (Table 17), yield of different biomass types of supply zones under each scenario (Table 18), nominal gasoline production of bioethanol demand zones (Table 19), road and rail distance between supply zones and potential biorefinery locations (Tables 20, 21), road and rail distance between potential biorefinery locations and demand zones (Tables 22, 23).

Results and discussion

The formulated models are coded in GAMS 24.8.5 and solved by CPLEX solver employing an Intel core i7 2.60 GHz laptop with 8 GB of RAM. The statistics of BSPP, RSPP and HWRSPP models are summarized in Table 3.

The optimal values of decision variables (X^*, X_n^* and Y^*) are achieved for BSPP (for different values of α, β and γ), RSPP (for different values of γ) and HWRSPP models. In order to validate the performance of the proposed models, a deterministic linear programming model called 'realization model' is developed. Realization model is a simulation of reality and is especially applied to evaluate the robustness and desirability of the proposed models. The performance of the proposed RSPP and HWRSPP models under nominal data are compared with the performance of the BSPP model for different values of α, β and γ. To do so, 20 random data sets are uniformly generated from the fuzzy range of imprecise parameters (f_{real}, g_{real}, h_{real}, l_{real}, $\text{ym}_{n\,\text{real}}$ and pg_{real}). Also, the optimal values of decision variables (X^*, X_n^* and Y^*) achieved from BSPP, RSPP and HWRSPP models are incorporated in the realization model (model 53). Newly defined variables, S_1 and S_2, are the decision variables of the realization model, which represent the amounts of violation from biomass supply and bioethanol demand constraints. Parameters δ_1 and δ_2 are unit penalty cost of such possible violations.

Table 4 The performance of the proposed models versus the BSPP model

Models	Parameters	Mean	SD	CV
BSPP(a)	$\gamma=0.00$, $\alpha,\beta=0.9$	7.379E+08	1.130E+07	0.0153
BSPP(b)	$\gamma=0.25$, $\alpha,\beta=0.8$	7.361E+08	1.123E+07	0.0153
BSPP(c)	$\gamma=0.50$, $\alpha,\beta=0.7$	7.327E+08	1.108E+07	0.0151
RSPP(a)	$\gamma=0.00$	7.385E+08	1.033E+07	0.0140
RSPP(b)	$\gamma=0.25$	7.381E+08	1.028E+07	0.0139
RSPP(c)	$\gamma=0.50$	7.373E+08	1.014E+07	0.0137
HWRSPP		7.395E+08	1.134E+07	0.0153

$$
\begin{aligned}
&\min Z_{\text{real}} = f_{\text{real}} \cdot Y^* + g_{\text{real}} \cdot X^* + h_{\text{real}} \cdot X_n^* - l_{\text{real}} \cdot X_n^* + \delta_1 S_1 + \delta_2 S_2 \\
&\text{s.t.} \\
&X_n^* \le A \cdot ym_{n\text{real}} + S_1 \\
&B \cdot X_n^* = 0 \\
&C \cdot Y^* \le X^* \le D \cdot Y^* \\
&E \cdot X_n^* \le X^* \\
&F \cdot Y^* \ge 0 \\
&G \cdot Y^* \le 1 \\
&X_n^* + S_2 \ge H \cdot \text{pg}_{\text{real}} \\
&K \cdot X_n^* \le U \\
&S_1, S_2 \ge 0
\end{aligned}
\tag{53}
$$

Table 5 The mean value of objective function of the selected model and its components ($)

Objective function value	Fixed opening costs	Variable opening costs	Harvest costs	Production costs	Import costs	Inventory holding costs	Transportation costs	Revenues from by-products
7.37E+08	4.47E+07	1.80E+08	2.81E+08	2.37E+08	4.36E+07	3.37E+05	6.73E+07	1.17E+08

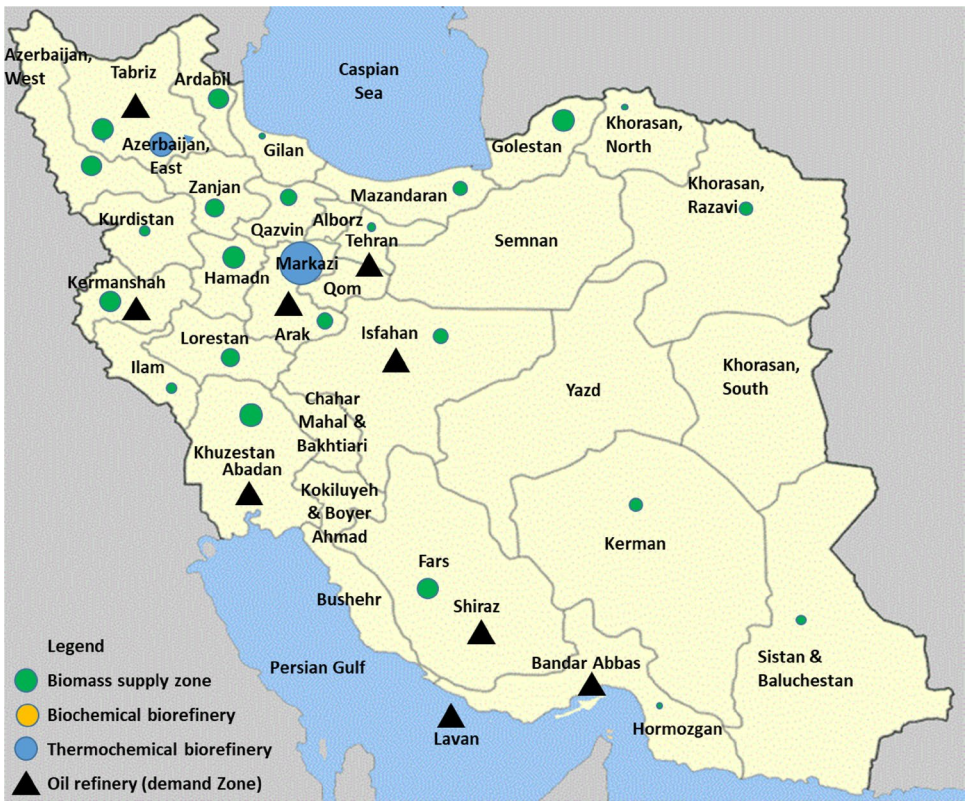


Fig. 2 Location of biorefineries of the selected model and biomass supply and bioethanol demand zones

Table 6 Production rate and inventory levels of each selected biorefinery

Biorefinery Location	Avg. annual production rate (gal/year)	Bioethanol inventory level (gal)	Avg. biomass inventory level (ton)	Max. biomass inventory level[a] (ton)	Min. biomass inventory level[b] (ton)
Azerbaijan, East	7.19E+7	0.00	2.38E+05	5.62E+05	0.00
Markazi	2.06E+8	0.00	5.37E+05	1.28E+06	0.00

[a]Maximum biomass inventory level is for summer

[b]Minimum biomass inventory level is for spring

A total of 140 runs, including 20 runs for each of 3 versions of BSPP and RSPP models (with different values for α, β and γ) and 20 runs for HWRSPP model, are performed in realization step. In order to assess the desirability and performance of the proposed models, the expected value of the objective function ($E[Z_{real}]$) and its standard deviation (SD) and coefficient of variation (CV) of RSPP and HWRSPP models are compared with the BSPP model and presented in Table 4. It is worth mentioning that none of the versions of the proposed model is the best for all problem cases, and the DMs can choose the one most meets their preferences and priorities. In this study, regarding the obtained results and DMs' priorities version (c) of RSPP model is selected.

Table 5 presents the mean value of the objective function of model RSPP(c) and its components. Except for inventory holding costs, all other components of the objective function show a major share in total value of the objective function.

Table 7 Bioethanol import rate of each oil refinery

No.	Oil refinery	Avg. annual import rate (gal/year)
1	Tabriz	0.00
2	Shiraz	0.00
3	Bandar Abbas	1.02E+06
4	Isfahan	0.00
5	Kermanshah	0.00
6	Abadan	4.23E+06
7	Arak	0.00
8	Tehran	0.00
9	Lavan	8.41E+06

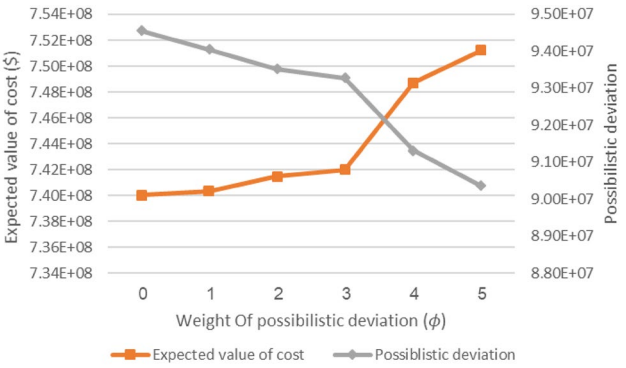


Fig. 3 Sensitivity analysis on weight of possibilistic deviation (ϕ)

Table 8 Biomass feedstock transported to each selected biorefinery

Biorefinery location	Supply zone (average annual amount of transported biomass in tons, transportation mode(s)[a])
Azerbaijan, East	Ardabil (2.69E+05, 1), Azerbaijan East (3.14E+05, 1,2), Azerbaijan West (2.75E+05, 2), Gilan (4.07E+03, 1), Qazvin (9.25E+03, 2), Zanjan (1.20E+05, 2)
Markazi	Fars (3.09E+05, 2), Golestan (3.52E+05, 2), Hamadan (3.51E+05, 1), Hormozgan (1.52E+03, 2), Ilam (2.70E+04, 1), Isfahan (9.61E+04, 2), Kerman (5.77E+04, 2), Kermanshah (3.20E+05, 1), Khorasan North (1.28E+03, 1), Khorasan Razavi (8.83E+04, 2), Khuzestan (3.91E+05, 2), Kurdistan (2.50E+04, 1), Lorestan (2.43E+05, 2), Markazi (1.73E+05, 1,2), Mazandaran (1.04E+05, 2), Qazvin (1.25E+05, 2), Sistan and Baluchestan (1.01E+04, 2), Tehran (4.83E+04, 2), Zanjan (8.80E+04, 2)

[a]1 and 2 refer to roadway and railway transportation, respectively

Table 9 Bioethanol transported from each selected biorefinery to each oil refinery

Biorefinery location	Oil refinery (average annual amount of transported bioethanol in gallons, transportation mode(s)[a])
Azerbaijan, East	Tabriz (1.59E+07, 1,2), Shiraz (6.42E+06, 2), Bandar Abbas (1.63E+07, 2), Isfahan (1.85E+06, 2), Kermanshah (2.98E+06, 1), Tehran (2.85E+07, 2)
Markazi	Shiraz (2.49E+06, 2), Bandar Abbas (2.48E+07, 2), Isfahan (5.41E+07, 1,2), Abadan (4.53E+07, 1), Arak (7.23E+07, 1,2), Tehran (4.19E+06, 1), Lavan (2.49E+06, 1)

[a]1 and 2 refer to roadway and railway transportation, respectively

The main purpose of the model is to determine the optimal values of strategic and tactical decision variables while considering operational and disruption risks. Selected location of biorefineries of RSPP(c) model together with biomass supply zones and bioethanol demand zones are shown in Fig. 2. Size of the circles is relatively correlated to their corresponding value. As can be seen, two thermochemical biorefineries are chosen to be placed in East Azerbaijan and Markazi provinces with annual capacity of 1.00E+06 and 2.78E+06 tons of biomass, respectively.

Due to space limitation, aggregated values of tactical decision variables are presented in Tables 6, 7, 8 and 9. Table 6 demonstrates the annual amount of bioethanol produced at each biorefinery as well as the inventory levels of biomass and bioethanol. It can be seen that bioethanol inventory level is zero. Also maximum biomass inventory level is more than twice the average inventory level while minimum biomass inventory level is zero. This is because biomass feedstock is harvested in harvest season (i.e. summer) and a large part of it is stored for future seasons to be processed. Also in the last season before the (next) harvest season, the inventory level becomes zero.

Table 7 presents average import amount of bioethanol to each oil refinery. It can be seen that bioethanol is imported to only three oil refineries, namely Bandar Abbas, Abadan and Lavan.

Table 8 shows the average amount of biomass feedstock transported from each supply zone to each selected

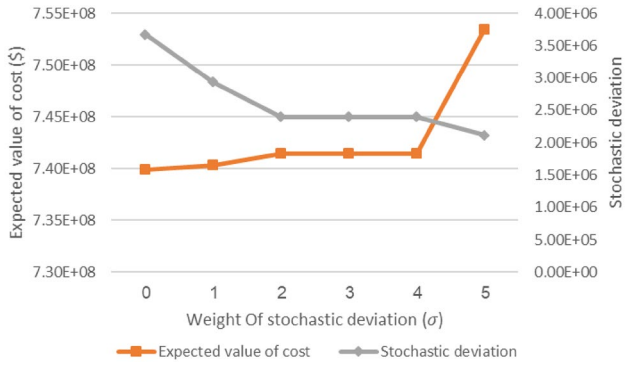


Fig. 4 Sensitivity analysis on weight of stochastic deviation (σ)

biorefinery. It also shows the selected transportation mode. In order to avoid prolongation, all four kinds of biomass have been summed up together. Table 9 displays the average amount of bioethanol transported from each selected biorefinery to each existing oil refinery as well as the selected transportation mode.

Sensitivity analysis

In this section, sensitivity analyses are done on some important parameters of the selected version of the RSPP model, i.e., RSPP(c), to see the impact of these parameters on the model. The parameters include weight of possibilistic deviation (ϕ), weight of stochastic deviation (σ), probability of occurrence of scenarios (p_n), blending percentage of bioethanol in gasoline (bf_t) and fixed setup cost of biorefineries ($\widetilde{fcr}_{j,s}$). The results of sensitivity analysis on parameters ϕ and σ are illustrated in Figs. 3 and 4. Figure 3 indicates that as the importance weight of possibilistic deviation (ϕ) in model (51) increases, the optimal expected value of cost increases and the optimal possibilistic deviation decreases. Likewise, Fig. 4 shows that by increasing the importance weight of stochastic deviation (σ), the optimal expected value of cost increases and the optimal stochastic deviation reduces.

The results of sensitivity analysis on probability of occurrence of scenarios (p_n) are demonstrated in Table 10. As mentioned before, p_1, p_2, p_3 and p_4 are probability of occurrence of no drought, mild, moderate and severe drought scenarios ($\Sigma p_n = 1$). Indeed, altering the values of p_n directly affects biomass supply of supply zones. Table 10 indicates that as the probability of occurrence of more severe drought scenarios increases, the average annual bioethanol production decreases and the average annual bioethanol import increases. In addition, it is obvious that the value of optimal objective function increases.

Table 10 Sensitivity analysis on probability of occurrence of scenarios (p_n)

Prob. of occurrence of scenarios				Objective function ($)	Biorefinery (annual capacity in tons of biomass)	Avg. annual bioethanol production (gal)	Avg. annual bioethanol import (gal)
p_1	p_2	p_3	p_4				
1	0	0	0	8.197E+08	Khuzestan (1.15E+06), Qazvin (2.74E+06)	2.835E+08	7.784E+06
0.5	0.5	0	0	8.238E+08	East Azerbaijan (1.12E+06), Markazi (2.74E+06)	2.835E+08	9.812E+06
0.5	0	0.5	0	8.337E+08	East Azerbaijan (1.06 E+06), Markazi (2.72E+06)	2.741E+08	1.714E+07
0.5	0	0	0.5	8.466E+08	East Azerbaijan (1.00E+06), Markazi (2.73E+06)	2.663E+08	2.500E+07
0	1	0	0	8.242E+08	East Azerbaijan (1.12E+06), Markazi (2.74E+06)	2.824E+08	8.931E+06
0	0.5	0.5	0	8.340E+08	East Azerbaijan (1.06 E+06), Markazi (2.72E+06)	2.755E+08	1.583E+07
0	0.5	0	0.5	8.469E+08	East Azerbaijan (1.00E+06), Markazi (2.73E+06)	2.674E+08	2.388E+07
0	0	1	0	8.349E+08	East Azerbaijan (1.06 E+06), Markazi (2.72E+06)	2.781E+08	1.315E+07
0	0	0.5	0.5	8.476E+08	East Azerbaijan (1.00E+06), Markazi (2.73E+06)	2.701E+08	2.117E+07
0	0	0	1	8.487E+08	East Azerbaijan (1.00E+06), Markazi (2.73E+06)	2.734E+08	1.791E+07

Table 11 Sensitivity analysis on blending percentage of bioethanol to gasoline (bf_t)

Parameter's value	Objective function ($)	Biorefinery (annual capacity in tons of biomass)	Average annual bioethanol production (gal)	Average annual bioethanol import (gal)
$0{\cdot}25 \times bf_t$	2.154E+08	Markazi (1.00E+06)	7.230E+7	5.229E+05
$0{\cdot}50 \times bf_t$	4.124E+08	Markazi (1.99E+06)	1.444E+08	1.278E+06
$0{\cdot}75 \times bf_t$	6.155E+08	Markazi (2.98E+06)	2.163E+08	2.195E+07
$1{\cdot}00 \times bf_t$	8.377E+08	East Azerbaijan (1.00E+06), Markazi (2.78E+06)	2.775E+08	1.380E+07
$1{\cdot}25 \times bf_t$	–	Infeasible	–	–

Table 12 Sensitivity analysis on fixed setup cost of biorefineries ($\widetilde{fcr}_{j,s}$)

Parameter's value	Objective function ($)	Biorefinery location (capacity in tons of biomass)	Avg. annual bioethanol production (gal)	Avg. annual bioethanol import (gal)
$0{\cdot}25\times\widetilde{fcr}_{j,s}$	7.959E+08	East Azerbaijan (1.00E+06), Fars (1.00E+06), Markazi (1.84E+06)	2.807E+08	1.063E+07
$0{\cdot}50\times\widetilde{fcr}_{j,s}$	8.131E+08	East Azerbaijan (1.00E+06), Markazi (2.78E+06)	2.775E+08	1.380E+07
$0{\cdot}75\times\widetilde{fcr}_{j,s}$	8.254E+08	East Azerbaijan (1.00E+06), Markazi (2.78E+06)	2.775E+08	1.380E+07
$1{\cdot}00\times\widetilde{fcr}_{j,s}$	8.377E+08	East Azerbaijan (1.00E+06), Markazi (2.78E+06)	2.775E+08	1.380E+07
$1{\cdot}25\times\widetilde{fcr}_{j,s}$	8.499E+08	East Azerbaijan (1.00E+06), Markazi (2.78E+06)	2.775E+08	1.380E+07
$1{\cdot}50\times\widetilde{fcr}_{j,s}$	8.622E+08	East Azerbaijan (1.00E+06), Markazi (2.78E+06)	2.775E+08	1.380E+07
$1{\cdot}75\times\widetilde{fcr}_{j,s}$	8.745E+08	East Azerbaijan (1.00E+06), Markazi (2.78E+06)	2.775E+08	1.380E+07
$2{\cdot}00\times\widetilde{fcr}_{j,s}$	8.867E+08	East Azerbaijan (1.00E+06), Markazi (2.78E+06)	2.775E+08	1.380E+07

Table 13 Sensitivity analysis on maximum annual allowed amount of carbon to be emitted (uc)

Parameter's value	Objective function ($)	Biorefinery (annual capacity in tons of biomass)	Avg. annual bioethanol production (gal)	Avg. annual bioethanol import (gal)
$0{\cdot}70\times$uc	–	Infeasible	–	–
$0{\cdot}80\times$uc	–	Infeasible	–	–
$0{\cdot}90\times$uc	8.548E+08	East Azerbaijan (1.00E+06), Markazi (2.52E+06)	2.525E+08	3.595E+07
$1{\cdot}00\times$uc	8.377E+08	East Azerbaijan (1.00E+06), Markazi (2.78E+06)	2.775E+08	1.380E+07
$1{\cdot}10\times$uc	8.357E+08	East Azerbaijan (1.00E+06), Markazi (2.98E+06)	2.883E+08	2.970E+06
$1{\cdot}20\times$uc	8.357E+08	East Azerbaijan (1.00E+06), Markazi (2.98E+06)	2.883E+08	2.970E+06
$1{\cdot}30\times$uc	8.357E+08	East Azerbaijan (1.00E+06), Markazi (2.98E+06)	2.883E+08	2.970E+06

The results of sensitivity analysis on blending percentage of bioethanol to gasoline (bf_t) are summarized in Table 11. Increasing the blending percentage of bioethanol actually increases the demand of bioethanol. Table 11 shows that by increasing the blending percentage of bioethanol, the sum of average annual bioethanol production and import increases to satisfy the demand. And it is obvious that the value of optimal objective function increases by increasing bf_t. When the blending percentage is multiplied by 1.25, the model becomes infeasible which shows that bioethanol production and import cannot meet the demand. This can be overcome by increasing the upper bound defined on bioethanol import (uim_t).

Table 12 shows sensitivity analysis on fixed setup cost of biorefineries ($\widetilde{fcr}_{j,s}$). It shows that optimal solution of the model remains the same for different values of $\widetilde{fcr}_{j,s}$ except for the first row. Actually when the fixed setup cost is small (the first row) the optimal solution is achieved by producing more bioethanol and having less imports. With the increase in fixed setup cost, bioethanol production is expected to fall and bioethanol import rise (the second row). But, this is not the case from the third row onwards, because of the upper bound on bioethanol import (uim_t). Again, it is obvious that increasing the value of $\widetilde{fcr}_{j,s}$ leads to increased value of the objective function.

Table 13 shows sensitivity analysis on maximum annual allowed amount of carbon to be emitted (uc), which is the right-hand side of constraint (17). Table 13 reveals that the model is infeasible up to $0{\cdot}80\times$uc. The model becomes feasible from $0{\cdot}90\times$uc onwards. By increasing the value of uc up to $1{\cdot}10\times$uc, the value of objective function decreases. Also the annual capacity of biorefinery located in Markazi province and the total capacity of biorefineries increase and consequently average annual bioethanol production rises and bioethanol import falls. From $1{\cdot}10\times$uc onwards, the values of all variables remain the same. This means that constraint (17) is active up to $1{\cdot}00\times$uc and afterward it becomes inactive.

Comparison analysis

In this section, the performance of the preferred RSPP(c) version of the model is compared with its counterpart model regardless drought scenarios, under nominal data. Actually,

Table 14 Comparison of resilient and non-resilient models performance

	Resilient model	Non-resilient model
Mean cost ($)	7.130E+08	7.160E+08
Standard deviation	1.084E+07	1.198E+07
Coefficient of variation	0.0152	0.0167
Biorefinery location (technology, annual capacity in tons of biomass)	East Azerbaijan (1, 1.00E+06), Markazi (1, 2.78E+06)	Khuzestan (1, 1.15E+06), Qazvin (1, 2.74E+06)
Total annual processing capacity of biorefineries (tons of biomass)	3.78E+06	3.89E+06
Average annual biomass consumption (ton)	3.73E+06	3.78E+06
Average annual bioethanol production (gallon)	2.76E+08	2.78E+08
Utilized percentage of biorefineries capacity (%)	98.68	97.17
Average annual bioethanol import (gallon)	7.80E+06	5.76E+06
Average biomass inventory level (ton)	7.42E+05	6.75E+05
Average bioethanol inventory level (gallon)	6.43E+02	9.85E+03
Average annual GHG emission (ton CO_2-equiv.)	1.08E+06	1.10E+06

RSPP(c) version of the model is both robust and resilient to the epistemic uncertainty of parameters and disruptions (called resilient model from now on), but its counterpart model is not robust and resilient to disruptions (called non-resilient model from now on). In both cases, the strategic decisions including biorefinery locations and capacities are fixed and only tactical decisions such as biomass flow and inventory levels can alter. This comparison makes it possible to assess the impact of ignoring disruption scenarios in the design phase on different factors, especially on the total cost. Some main results of resilient and non-resilient model are summarized in Table 14.

Managerial implications

Comparing the main results of the resilient and non-resilient models, some useful managerial implications can be drawn as follows:

- Mean cost and its coefficient of variation have less values in the proposed resilient model.
- The location of biorefineries differs in the resilient and non-resilient models.
- Total processing capacity of biorefineries (which is sum of capacities of all installed refineries presented in the fifth row) has a greater value in the non-resilient model.
- The average annual biomass consumption and consequently the average annual bioethanol production are slightly more in the non-resilient model.
- Utilization percentage of biorefineries has a less value in the non-resilient model. Utilization percentage of biorefineries is calculated through dividing average annual biomass consumption at refineries by total annual processing capacity of the biorefineries.
- The average value of annual bioethanol import is greater in the resilient model, showing that bioethanol import is economical against the fluctuations caused by drought. But, since the non-resilient model does not account for these fluctuations it installs more capacity of biorefineries.
- Average biomass inventory level is more in the resilient model, whereas average bioethanol inventory level is more in the non-resilient model. However, bioethanol inventory level is very low in both models compared to biomass inventory level.
- Average amount of annual GHG emission is less in the resilient model.

Conclusions and future research

An MILP model for the design and planning of an LBSC is developed in this study. The proposed model determines both strategic and tactical decisions by optimizing the economic objective function under epistemic uncertainties and disruptions of model parameters. Strategic decisions include facilities location, capacity and technology type, and tactical decisions include biomass allocation, biofuel production or import quantities, material inventory levels and material transportation amounts. Different costs, market prices, biomass supply and biofuel demand are sources of epistemic uncertainty. Biomass supply is also affected by a natural disruption, namely drought. A hybrid robust stochastic-possibilistic programming (RSPP) approach is employed to deal with the above-mentioned uncertainties and disruption. This approach applies a combination of possibilistic and stochastic approaches and robust optimization. The recently introduced fuzzy *Me* measure is used in PCCP to

provide flexibility in decision-making. In addition to RSPP approach, a hard worst-case robust stochastic-possibilistic programming (HWRSPP) approach is proposed to capture the above-mentioned uncertainties and disruption. As far as we know, this is the first research which applies RSPP and HWRSPP approaches for the design of a BSCN.

The proposed RSPP and HWRSPP models are applied in an Iranian case study to design a BSCN. The RSPP(c) version of the model is selected regarding the obtained results and preferences of the DMs. In accordance with this version, East Azerbaijan and Markazi provinces are chosen as the best locations for setting up two thermochemical biorefineries with annual processing capacity of 1.00E+06 and 2.78E+06 tons of biomass, respectively. Also, this version of the model (which is both robust against and resilient to drought disruption in the supply of biomass) is compared with its counterpart model which does not consider drought disruption. The results reveal the superiority of RSPP(c) model over its non-resilient counterpart both in cost and GHG emissions.

The proposed model can be expanded to consider environmental and social objective functions as well as the economic one. Another extension could be incorporating different types of uncertainty including deep uncertainties, epistemic and randomness. Also, considering other existing disruptions in BSCN design models can make them more realistic. The mentioned expansions can lead to more complexities in BSCN design models, and it may be necessary to apply decomposition methods and modified heuristics/metaheuristics to solve the model in an acceptable processing time.

Appendix

Tables 15, 16, 17, 18, 19, 20, 21, 22 and 23.

Table 15 Key deterministic parameters

Symbol	Value	References
MC_b	$MC_1=0.12$, $MC_2=MC_3=MC_4=0.15$	Bairamzadeh et al. (2018)
TCD_m	$TCD_1=28$, $TCD_2=121$	Xie et al. (2014)
TCL_m	$TCL_1=8000$, $TCL_2=33{,}000$	Xie et al. (2014)
ν	$\nu=0.3$	Osmani and Zhang (2013)
$\omega_{b,i,t}$	$\omega_{b,i,t}=0.05$ (for all b, i, t)	Čuček et al. (2014)
$\varphi_{b,s}$	$\varphi_{1,1}=72$, $\varphi_{1,2}=75$ $\varphi_{2,1}=72$, $\varphi_{2,2}=60$ $\varphi_{3,1}=82$, $\varphi_{3,2}=75$ $\varphi_{4,1}=66$, $\varphi_{4,2}=72$	Bairamzadeh et al. (2018)
$\lambda_{b,s,c}$	$\lambda_{1,1,1}=12.85$, $\lambda_{1,2,2}=0.186$ $\lambda_{2,1,1}=13.42$, $\lambda_{2,2,2}=0.219$ $\lambda_{3,1,1}=13.42$, $\lambda_{3,2,2}=0.219$ $\lambda_{4,1,1}=13.42$, $\lambda_{4,2,2}=0.219$ $\lambda_{b,1,2}=0$ (for all b) $\lambda_{b,2,1}=0$ (for all b)	Bairamzadeh et al. (2018)
LR_s	$LR_s=1{,}000{,}000$ (for all s)	Osmani and Zhang (2014)
UR_s	$UR_s=3{,}000{,}000$ (for all s)	Osmani and Zhang (2014)
ψ	$\psi=0.8$	
BF_t	$BF_t=0.05$ (for all t)	National Document for Knowledge-based Development of Renewable energies (2015)
$OBF_{b,i}$	$OBF_{b,i}=0.2$ (for all b, i)	
CHM_b	$CHM_b=0.175$ (for all b)	Osmani and Zhang (2014)
$CPM_{b,s}$	$CPM_{b,1}=0.09$, $CPM_{b,2}=0.08$ (for all b)	Osmani and Zhang (2014)
$CTM_{b,m}$	$CTM_{b,1}=6.84E{-}05$, $CTM_{b,2}=1.84E{-}05$ (for all b)	Osmani and Zhang (2014)
CTE_m	$CTE_1=2.05E{-}07$, $CTE_2=5.51E{-}08$	Osmani and Zhang (2014)

*$b=1$ refers to corn stover, $b=2$ refers to wheat straw, $b=3$ refers to barley straw and $b=4$ refers to rice straw

*$s=1$ refers to thermochemical and $s=2$ refers to biochemical conversion technology

*$c=1$ refers to mixed alcohol and $c=2$ refers to electricity

*$m=1$ refers to road and $m=2$ refers to rail transportation

Table 16 Nominal values of uncertain parameters

Symbol	Value	References
$bp_{c,t}$	$bp_{1,t}=2.3$, $bp_{2,t}=85$ (for all t)	Osmani and Zhang (2014)
imp_t	$imp_t=3.2$ (for all t)	
$fcr_{j,s}$	$fcr_{j,1}=5{,}575{,}000$, $fcr_{j,2}=5{,}325{,}000$ (for all j)	Osmani and Zhang (2014)
$vcr_{j,s}$	$vcr_{j,1}=48$, $vcr_{j,2}=46$ (for all j)	Osmani and Zhang (2014)
$pce_{j,s}$	$pce_{j,1}=0.859$, $pce_{j,2}=0.907$ (for all j)	Osmani and Zhang (2014)
$hcm_{b,i}$	$hcm_{b,i}=70$ (for all b, i)	
$ice_{j,t}$	$ice_{j,t}=0.06$ (for all j, t)	Bairamzadeh et al. (2015)
$icm_{b,j,t}$	$icm_{b,j,t}=0.1125$ (for all b, j, t)	Bairamzadeh et al. (2015)
$tcm_{b,i,j,m}$	$tcm_{b,i,j,1}=1.108$, $tcm_{b,i,j,2}=1.562$ (for all b, i, j)	Bairamzadeh et al. (2015); Xie et al. (2014)
$lcm_{b,i,j,m}$	$lcm_{b,i,j,1}=112$, $lcm_{b,i,j,2}=484$ (for all b, i, j)	Bairamzadeh et al. (2015); Xie et al. (2014)
$tce_{j,k,m}$	$tce_{j,k,1}=1.2077$, $tce_{j,k,2}=1.562$ (for all j, k)	Bairamzadeh et al. (2015); Xie et al. (2014)
$lce_{j,k,m}$	$lce_{j,k,1}=160$, $lce_{j,k,2}=495$ (for all j, k)	Bairamzadeh et al. (2015); Xie et al. (2014)
$ym_{b,t}$	$ym_{1,t}=4.4$, $ym_{2,t}=3.5$, $ym_{3,t}=2.76$, $ym_{4,t}=4.9$ (for all t)	Bairamzadeh et al. (2018)

Table 17 Allocated land area of supply zones for cultivation of biomass feedstocks (ha)

i	Supply zone	$l_{i,b}$			
		Corn stover	Wheat straw	Barley straw	Rice straw
1	Ardabil	12,701	334,888	123,231	517
2	Azerbaijan, East	495	385,785	82,738	2187
3	Azerbaijan, West	5606	350,849	49,949	30
4	Fars	27,586	408,049	133,112	34,642
5	Gilan	32	8180	6797	151,949
6	Golestan	964	370,283	66,043	47,179
7	Hamadan	5813	414,238	110,144	0
8	Hormozgān	6060	11,996	992	0
9	Ilam	11,165	140,444	44,730	2985
10	Isfahan	1209	88,180	55,872	4933
11	Kerman	29,062	85,874	25,476	0
12	Kermanshah	33,907	431,845	164,000	258
13	Khorasan, North	0	163,525	57,441	1452
14	Khorasan, Razavi	0	267,858	175,001	1964
15	Khuzestan	78,317	511,965	116,309	73,104
16	Kurdistan	3034	573,655	29,410	0
17	Lorestan	2155	247,697	121,342	3666
18	Markazi	54	210,002	51,348	0
19	Mazandaran	75	55,993	23,747	199,366
20	Qazvin	4379	153,689	42,105	2323
21	Sistan and Baluchestan	5710	85,318	16,394	2758
22	Tehran	0	41,001	36,560	0
23	Zanjan	0	321,461	30,520	689

Iran Statistical Yearbook of Agricultural Products (2013)

Table 18 Yield of different biomass types of each supply zone in each planning period and under each scenario

Supply zone	Biomass type	$ymd_{b,i,t,n}$			
		$n=1$	$n=2$	$n=3$	$n=4$
		$t=1, 2, 3, 4$	$t=1, 2, 3, 4$	$t=1, 2, 3, 4$	$t=1, 2, 3, 4$
$i=1, 2, 3, 4, 7, 8, 9, 10, 12, 16, 17, 18, 20, 22$	$b=1$	0, (4,00, 4.20, 4.50, 5.00), 0, 0	0, (3.60, 3.78, 4.05, 4.50), 0, 0	0, (2.80, 2.94, 3.15, 3.50), 0, 0	0, (2.40, 2.52, 2.70, 3.00), 0, 0
	$b=2$	(3.10, 3.30, 3.50, 4.00), 0, 0, 0	(2.79, 2.97, 3.15, 3.60), 0, 0, 0	(2.17, 2.31, 2.45, 2.80), 0, 0, 0	(1.86, 1.98, 2.10, 2.40), 0, 0, 0
	$b=3$	0, 0, 0, (2.50, 2.60, 2.80, 3.00)	0, 0, 0, (2.25, 2.34, 2.52, 2.70)	0, 0, 0, (1.75, 1.82, 1.96, 2.10)	0, 0, 0, (1.50, 1.56, 1.68, 1.80)
	$b=4$	0, (4.30, 4.70, 4.80, 5.60), 0, 0	0, (3.87, 4.23, 4.32, 5.04), 0, 0	0, (3.01, 3.29, 3.36, 3.92), 0, 0	0, (2.58, 2.82, 2.88, 3.36), 0, 0
$i=5, 6, 19$	$b=1$	0, (4,00, 4.20, 4.50, 5.00), 0, 0	0, (4,00, 4.20, 4.50, 5.00), 0, 0	0, (3.60, 3.78, 4.05, 4.50), 0, 0	0, (2.80, 2.94, 3.15, 3.50), 0, 0
	$b=2$	(3.10, 3.30, 3.50, 4.00), 0, 0, 0	(3.10, 3.30, 3.50, 4.00), 0, 0, 0	(2.79, 2.97, 3.15, 3.60), 0, 0, 0	(2.17, 2.31, 2.45, 2.80), 0, 0, 0
	$b=3$	0, 0, 0, (2.50, 2.60, 2.80, 3.00)	0, 0, 0, (2.50, 2.60, 2.80, 3.00)	0, 0, 0, (2.25, 2.34, 2.52, 2.70)	0, 0, 0, (1.75, 1.82, 1.96, 2.10)
	$b=4$	0, (4.30, 4.70, 4.80, 5.60), 0, 0	0, (4.30, 4.70, 4.80, 5.60), 0, 0	0, (3.87, 4.23, 4.32, 5.04), 0, 0	0, (3.01, 3.29, 3.36, 3.92), 0, 0
$i=11, 13, 15$	$b=1$	0, (4,00, 4.20, 4.50, 5.00), 0, 0	0, (2.80, 2.94, 3.15, 3.50), 0, 0	0, (2.00, 2.10, 2.25, 2.50), 0, 0	0, (1.20, 1.26, 1.35, 1.50), 0, 0
	$b=2$	(3.10, 3.30, 3.50, 4.00), 0, 0, 0	(2.17, 2.31, 2.45, 2.80), 0, 0, 0	(1.55, 1.65, 1.75, 2.00), 0, 0, 0	(0.93, 0.99, 1.05, 2.00), 0, 0, 0
	$b=3$	0, 0, 0, (2.50, 2.60, 2.80, 3.00)	0, 0, 0, (1.75, 1.82, 1.96, 2.10)	0, 0, 0, (1.25, 1.30, 1.40, 1.50)	0, 0, 0, (0.75, 0.78, 0.84, 0.90)
	$b=4$	0, (4.30, 4.70, 4.80, 5.60), 0, 0	0, (3.01, 3.29, 3.36, 3.92), 0, 0	0, (2.15, 2.35, 2.40, 2.80), 0, 0	0, (1.29, 1.41, 1.44, 1.68), 0, 0
$i=14, 21, 23$	$b=1$	0, (4,00, 4.20, 4.50, 5.00), 0, 0	0, (3.20, 3.36, 3.60, 4.00), 0, 0	0, (2.40, 2.52, 2.70, 3.00), 0, 0	0, (2.00, 2.10, 2.25, 2.50), 0, 0
	$b=2$	(3.10, 3.30, 3.50, 4.00), 0, 0, 0	(2.48, 2.64, 2.80, 3.20), 0, 0, 0	(1.86, 1.98, 2.10, 2.40), 0, 0, 0	(1.55, 1.65, 1.75, 2.00), 0, 0, 0
	$b=3$	0, 0, 0, (2.50, 2.60, 2.80, 3.00)	0, 0, 0, (2.00, 2.08, 2.24, 2.40)	0, 0, 0, (1.50, 1.56, 1.68, 1.80)	0, 0, 0, (1.25, 1.30, 1.40, 1.50)
	$b=4$	0, (4.30, 4.70, 4.80, 5.60), 0, 0	0, (3.44, 3.76, 3.84, 4.48), 0, 0	0, (2.58, 2.82, 2.88, 3.36), 0, 0	0, (2.15, 2.35, 2.40, 2.80), 0, 0

Table 19 Gasoline production of bioethanol demand zones (gal/season)

k	Demand zone	Gasoline production of oil refinery k (for all t)			
		$pg_{k,1}$	$pg_{k,2}$	$pg_{k,3}$	$pg_{k,4}$
1	Tabriz	78,596,160	76,060,800	76,060,800	78,596,160
2	Shiraz	44,210,340	42,784,200	42,784,200	44,210,340
3	Bandar Abbas	208,771,050	202,036,500	202,036,500	208,771,050
4	Isfahan	277,542,690	268,589,700	268,589,700	277,542,690
5	Kermanshah	14,736,780	14,261,400	14,261,400	14,736,780
6	Abadan	245,613,000	237,690,000	237,690,000	245,613,000
7	Arak	358,594,980	347,027,400	347,027,400	358,594,980
8	Tehran	162,104,580	156,875,400	156,875,400	162,104,580
9	Lavan	54,034,860	52,291,800	52,291,800	54,034,860

Iran Statistical Yearbook of Oil Products Consumption (2014)

Table 20 Road distance between supply zones and potential biorefinery locations (km)

Supply zone *i*	Potential biorefinery location j													
	Azerbaijan, East	Fars	Hamadan	Hormozgān	Isfahan	Kerman	Kermanshah	Khorasan, Razavi	Khuzestan	Markazi	Mazandaran	Qazvin	Tehran	Zanjan
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	219	1515	667	1925	1030	1629	791	1333	1305	843	634	451	591	377
2	0	1523	609	1933	1038	1637	588	1493	1075	785	866	455	599	280
3	308	1559	610	2026	1074	1735	582	1801	1064	786	1174	763	907	588
4	1523	0	949	619	485	571	1112	1374	659	773	1191	965	924	1243
5	485	1249	401	1659	764	1361	590	1067	1039	577	368	185	325	348
6	996	1321	734	1731	836	1435	923	569	1281	690	130	547	397	716
7	609	949	0	1561	479	1179	189	1231	638	176	604	244	337	329
8	1933	619	1561	0	975	485	1769	1374	1278	1245	1601	1455	1334	1653
9	772	1100	397	1729	678	766	184	1604	447	514	977	617	710	598
10	1038	485	479	975	0	661	653	1222	745	288	706	480	439	757
11	1637	571	1179	485	661	0	1288	889	1230	949	1305	1182	1038	1357
12	588	1112	189	1769	653	1288	0	1420	487	365	793	433	526	414
13	1312	1637	1050	1627	1152	1142	1239	253	1587	1006	446	863	713	1032
14	1493	1374	1231	1374	1222	889	1420	0	1768	1187	699	1044	894	1213
15	1075	659	638	1278	745	1230	487	1768	0	581	1141	882	874	967
16	1337	174	763	793	299	745	952	1454	433	587	1005	779	738	1057
17	879	855	263	1327	370	1031	320	1393	375	206	766	507	499	592
18	785	773	176	1245	288	949	365	1187	581	0	560	303	239	505
19	866	1191	604	1601	706	1305	793	699	1141	560	0	417	267	586
20	455	965	244	1455	480	1182	433	1044	882	303	417	0	150	175
21	2166	1100	1654	734	1190	529	1817	951	1759	1478	1650	1717	1567	1886
22	599	924	337	1334	439	1038	526	894	874	239	267	150	0	319
23	280	1243	329	1653	757	1357	414	1213	967	505	586	175	319	0

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Table 21 Rail distance between supply zones and potential biorefinery locations (km)

Supply zone *i*	Potential biorefinery location *j*													
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	–	–	–	–	–	–	–	–	–	–	–	–	–	–
2	0	1800	–	2110	1272	1733	–	1667	1540	1051	1095	592	736	421
3	151	1951	–	2261	1423	1908	–	1821	1693	1195	1246	743	889	572
4	1800	0	–	1543	528	1190	–	1999	1532	1043	1427	1208	1078	1379
5	–	–	–	–	–	–	–	–	–	–	–	–	–	–
6	1245	1570	–	1879	1041	1526	–	1197	1311	820	142	645	498	816
7	–	–	–	–	–	–	–	–	–	–	–	–	–	–
8	2110	1543	–	0	1015	873	–	1492	1844	1353	1737	1518	1388	1689
9	–	–	–	–	–	–	–	–	–	–	–	–	–	–
10	1272	528	–	1015	0	662	–	1281	1006	515	899	680	548	851
11	1733	1190	–	873	662	0	–	1112	1491	1000	1384	1165	1035	1336
12	–	–	–	–	–	–	–	–	–	–	–	–	–	–
13	–	–	–	–	–	–	–	–	–	–	–	–	–	–
14	1667	1999	–	1492	1281	1112	–	0	1739	1250	1052	1075	928	1246
15	1540	1532	–	1844	1006	1491	–	1739	0	503	1167	948	818	1119
16	–	–	–	–	–	–	–	–	–	–	–	–	–	–
17	1191	1183	–	1493	655	1140	–	1390	351	147	818	599	469	770
18	1051	1043	–	1353	515	1000	–	1250	503	0	671	452	322	623
19	1095	1427	–	1737	899	1384	–	1052	1167	671	0	503	356	674
20	592	1208	–	1518	680	1165	–	1075	948	452	503	0	144	171
21	2379	1812	–	1495	1284	670	–	1758	2111	1615	2006	1787	1655	1958
22	736	1078	–	1388	548	1035	–	928	818	322	356	144	0	317
23	421	1379	–	1689	851	1336	–	1246	1119	623	674	171	317	0

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Table 22 Road distance between potential biorefinery location*s* and demand zones (km)

Potential biorefinery location *j*	Demand zone (oil refinery) *k*								
	Tabriz	Shiraz	Bandar Abbas	Isfahan	Kermanshah	Abadan	Arak	Tehran	Lavan
1	0	1523	1933	1038	588	1075	785	599	2300
2	1523	0	619	485	1112	659	773	924	986
3	609	949	1561	479	189	638	176	337	1928
4	1933	619	0	975	1769	1278	1245	1334	367
5	1038	485	975	0	653	745	288	439	1342
6	1637	571	485	661	1288	1230	949	1038	852
7	588	1112	1769	653	0	487	365	526	2136
8	1493	1374	1374	1222	1420	1768	1187	894	1741
9	1075	659	1278	745	487	0	581	874	1584
10	785	773	1245	288	365	581	0	239	1612
11	866	1191	1601	706	793	1141	560	267	1968
12	455	965	1455	480	433	882	303	150	1822
13	599	924	1334	439	526	874	239	0	1701
14	280	1243	1653	757	414	967	505	319	2020

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Table 23 Rail distance between potential biorefinery locations and demand zones (km)

Potential biorefinery location *j*	Demand zone (oil refinery) *k*								
	Tabriz	Shiraz	Bandar Abbas	Isfahan	Kermanshah	Abadan	Arak	Tehran	Lavan
1	0	1800	2110	1272	–	–	1051	736	–
2	1800	0	1543	528	–	–	1043	1078	–
3	–	–	–	–	–	–	–	–	–
4	2110	1543	0	1015	–	–	1353	1388	–
5	1272	528	1015	0	–	–	515	548	–
6	1733	1190	873	662	–	–	1000	1035	–
7	–	–	–	–	–	–	–	–	–
8	1667	1999	1492	1281	–	–	1250	928	–
9	1540	1532	1844	1006	–	–	503	818	–
10	1051	1043	1353	515	–	–	0	322	–
11	1095	1427	1737	899	–	–	671	356	–
12	592	1208	1518	680	–	–	452	144	–
13	736	1078	1388	548	–	–	322	0	–
14	421	1379	1689	851	–	–	623	317	–

Asia Seir Aras Company (www.asiaseiraras.com)

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